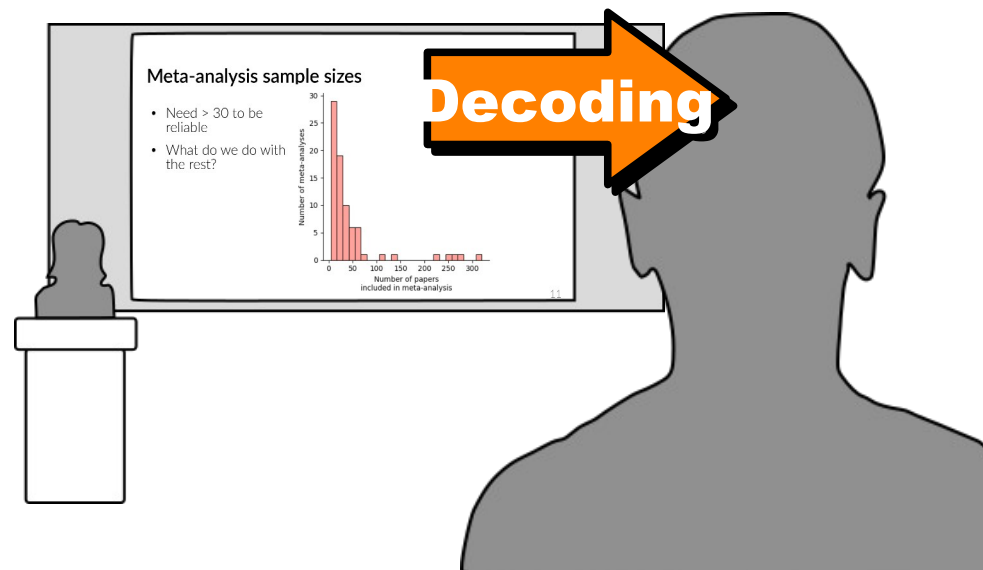




# Introduction to Data Visualization

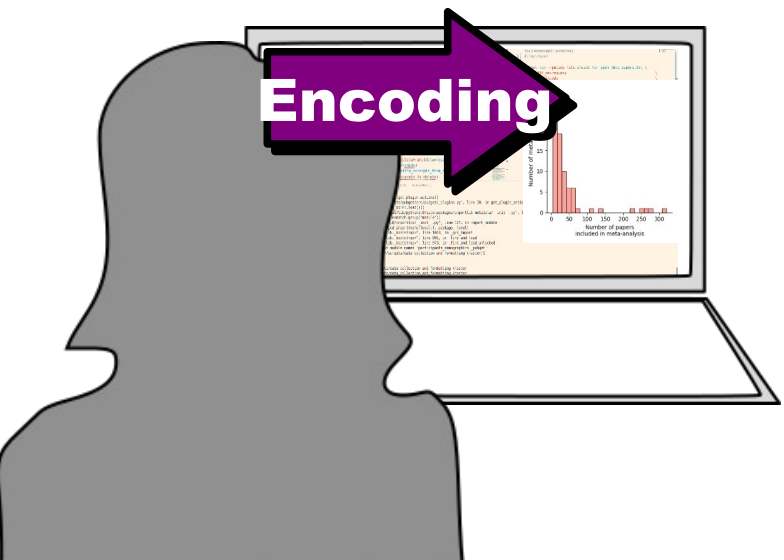
## Part 1: Decoding

Johanna Bayer  
Slides by Kendra Oudyk

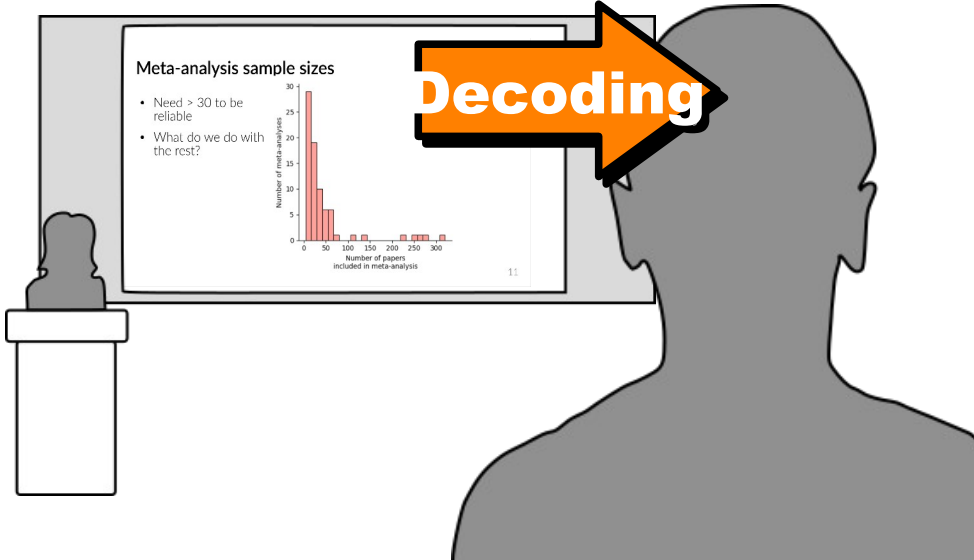


# Goal

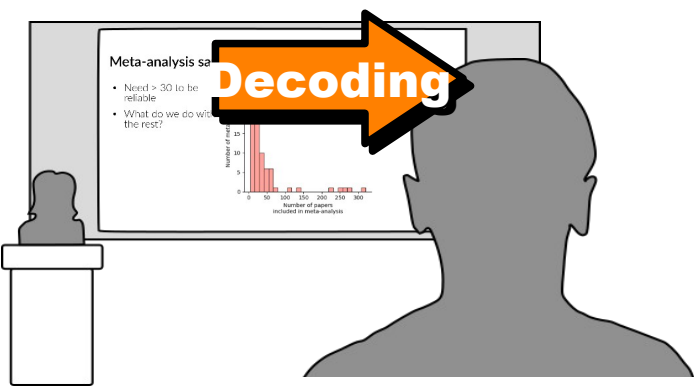
Use principles of visual **encoding** and **decoding** to **efficiently** create visualizations that are **effective** and **reproducible**



Encoding



Decoding



To plan an **effective visualization**, we need to think about

- **Message**

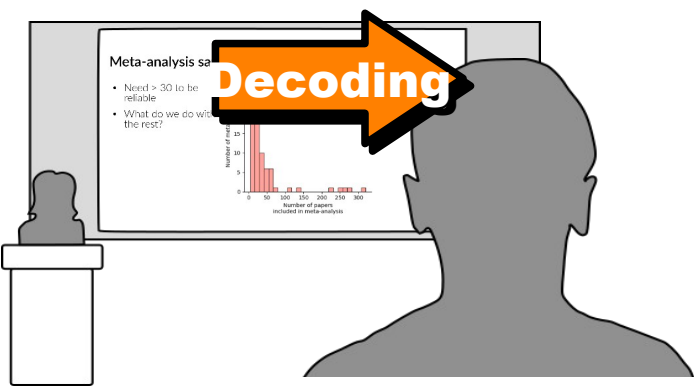
- **What** we want to communicate

- **Perception**

- **How best** to communicate it

- **Conventions**

- **How** it's **usually** communicated



To plan an **effective visualization**, we need to think about

- **Message**

- **What** we want to communicate

- **Perception**

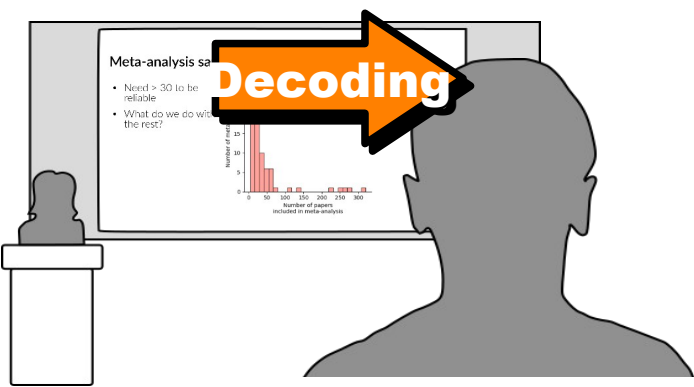
- **How best** to communicate it

- **Conventions**

- **How** it's **usually** communicated

# Message

- Raw data has no message
- Abstract it
  - “There are more males than females in science”  
--> A difference between magnitudes
  - “These brain areas activate together”  
--> A grouping / pattern



To plan an **effective visualization**, we need to think about

- **Message**

- **What** we want to communicate

- **Perception**

- **How best** to communicate it

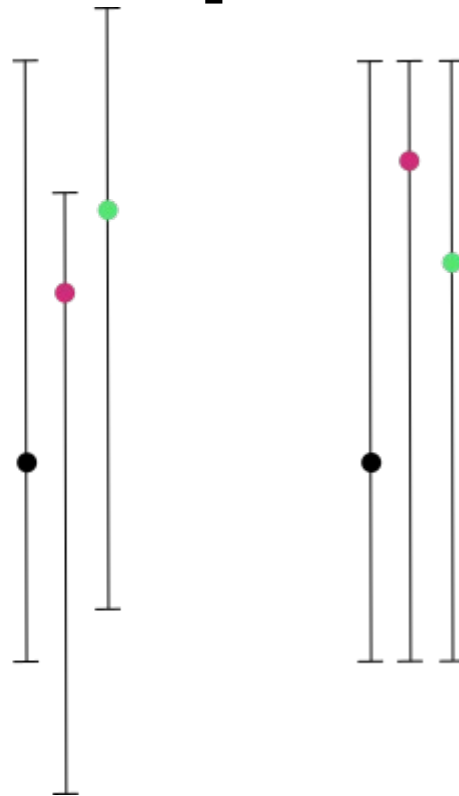
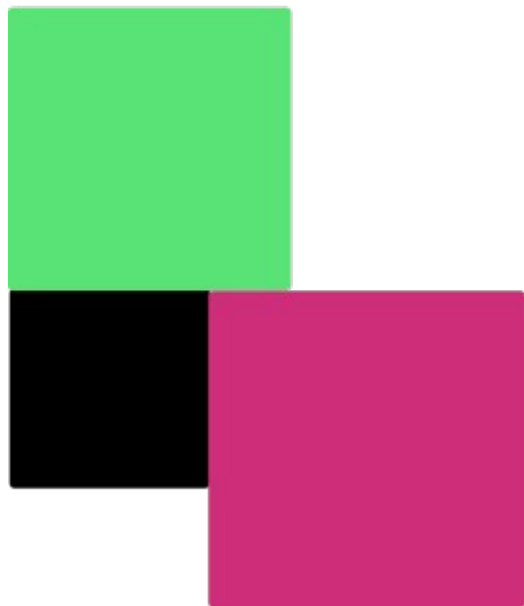
- **Conventions**

- **How** it's **usually** communicated

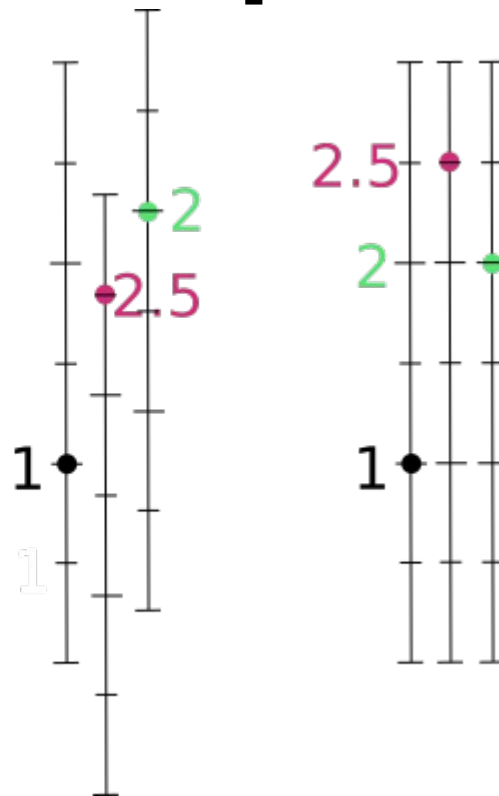
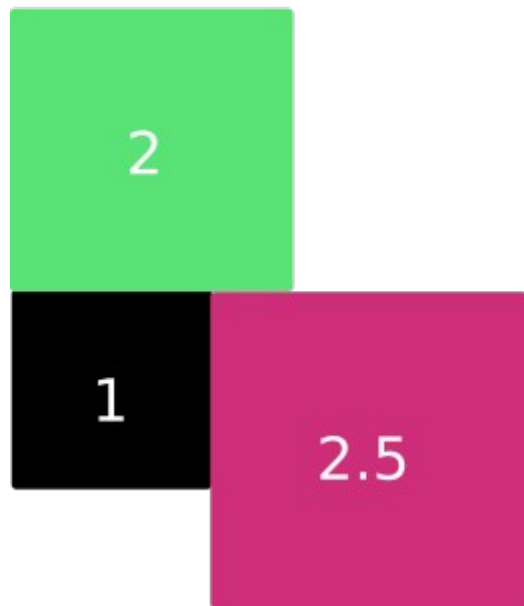
# Which blue bar is bigger?



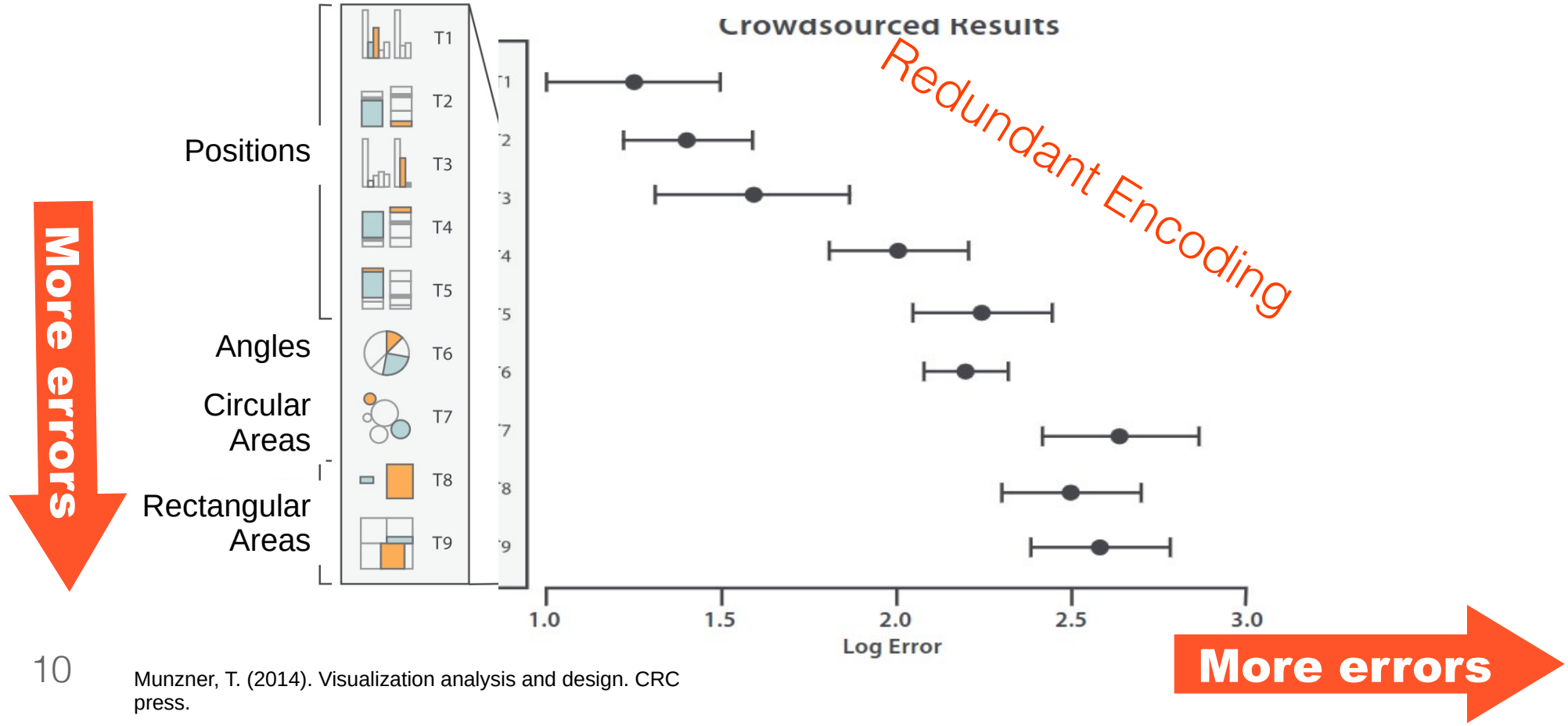
# Which is 2x black, green or pink?



# Which is 2x black, green or pink?



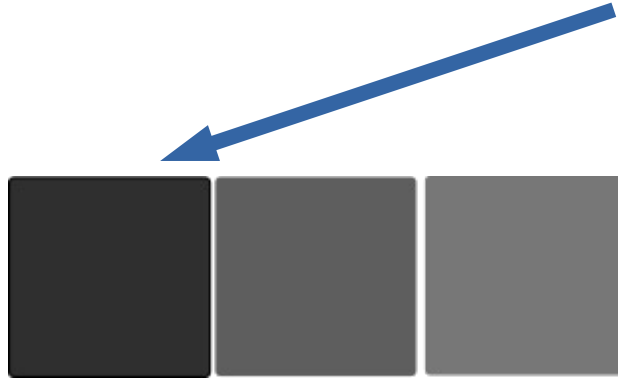
# We're better at judging aligned positions



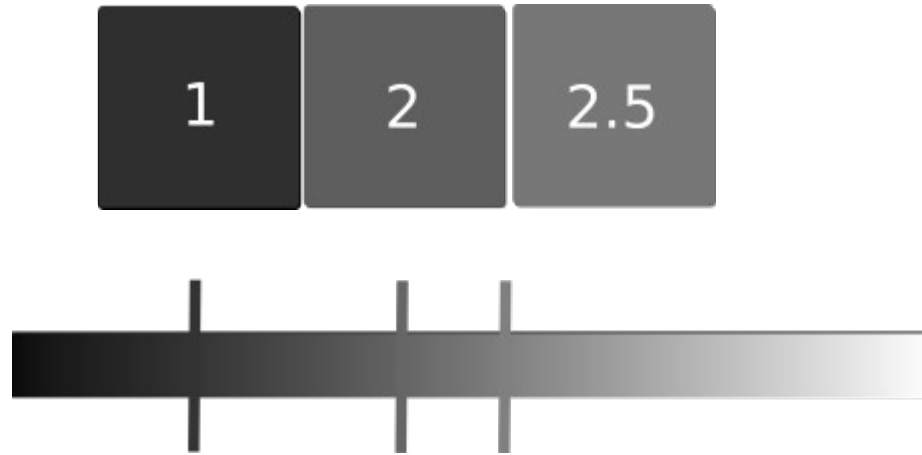
# Decoding values vs. patterns

- “There are more males than females in science”  
--> A difference between **magnitudes**
- “These brain areas activate together”  
--> A grouping / **pattern**

**Which is 2x lighter than the leftmost box?**

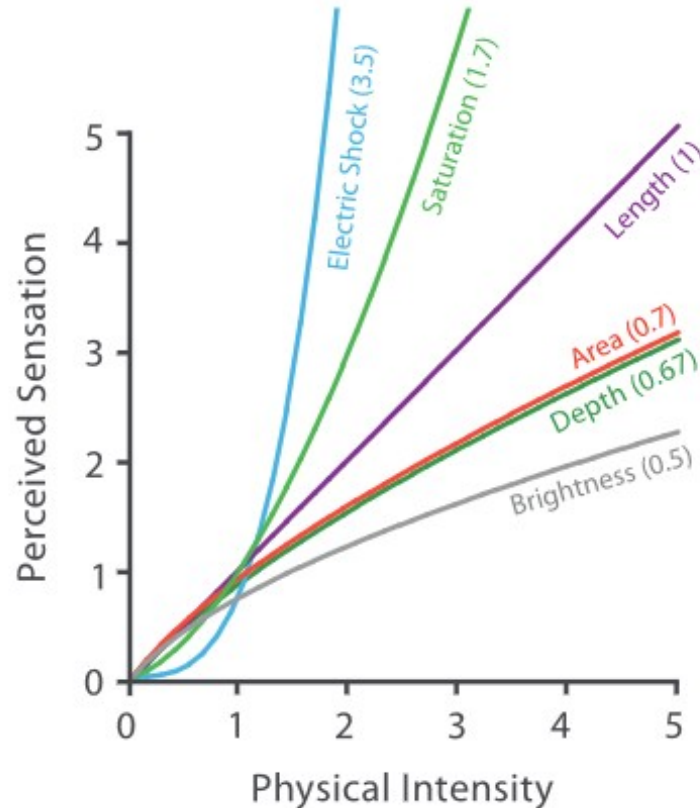


**Which is 2x lighter than the leftmost box?**



# Physical intensity vs perceived intensity

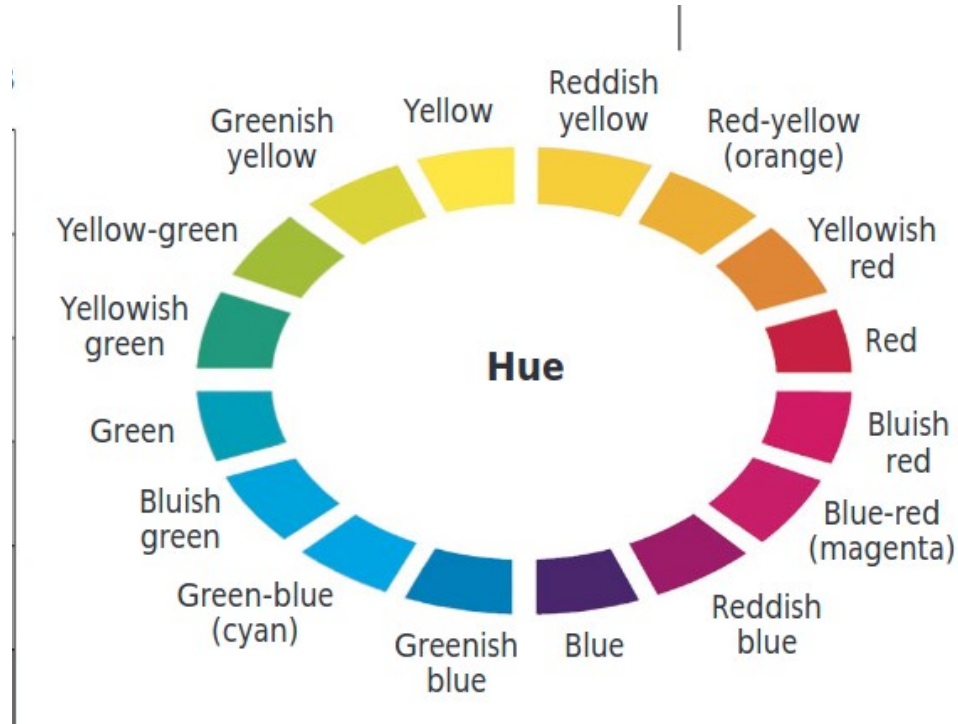
Steven's Psychophysical Power Law:  $S = I^N$



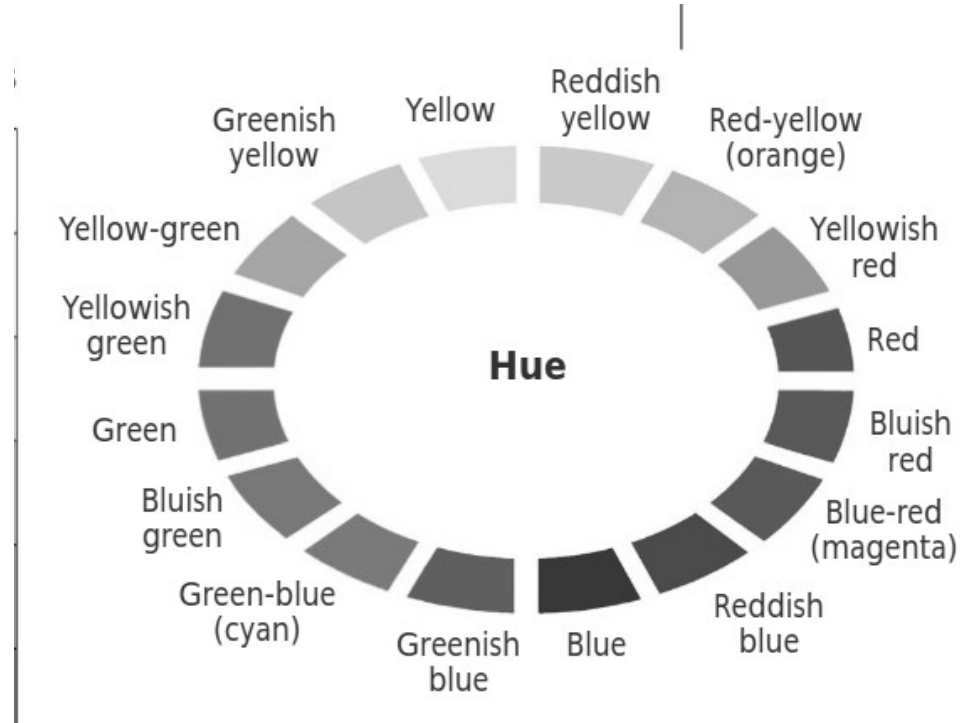
# Color: salient but complicated



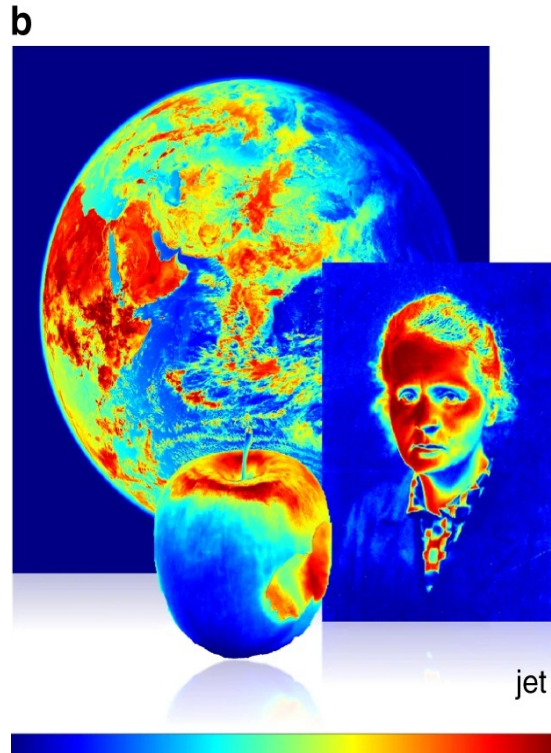
# Hue: how we talk about color



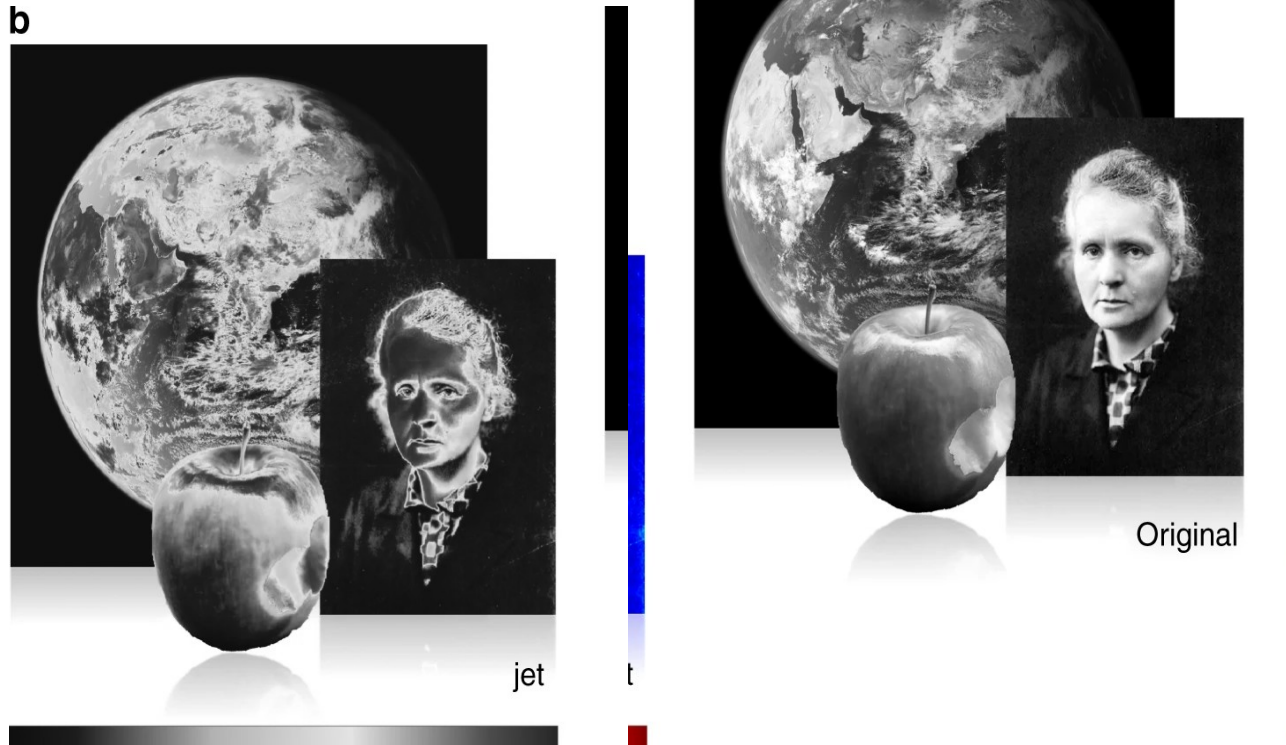
# Luminance: an important subconscious cue



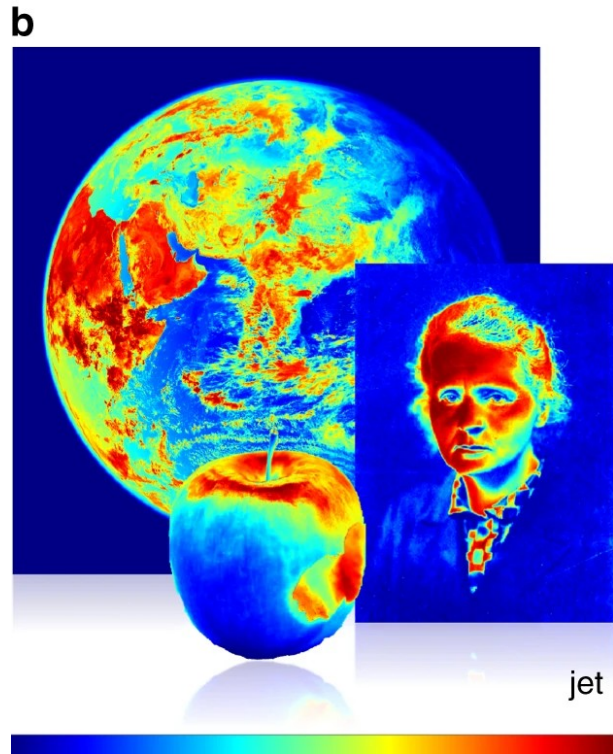
# Named colors don't work well for ordered values



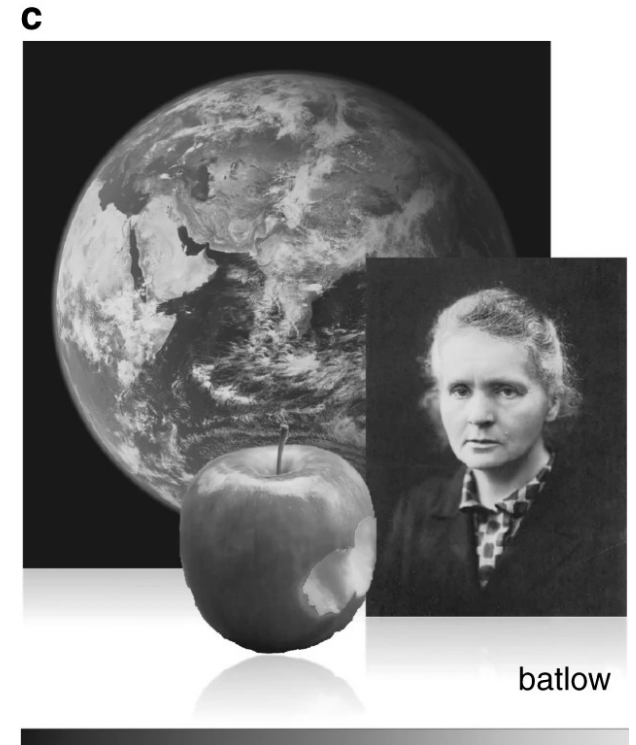
# Named colors don't work well for ordered values



# Certain colormaps do work



# Certain colormaps do work



Which is  
lightest?



**Physical**  
Lightness



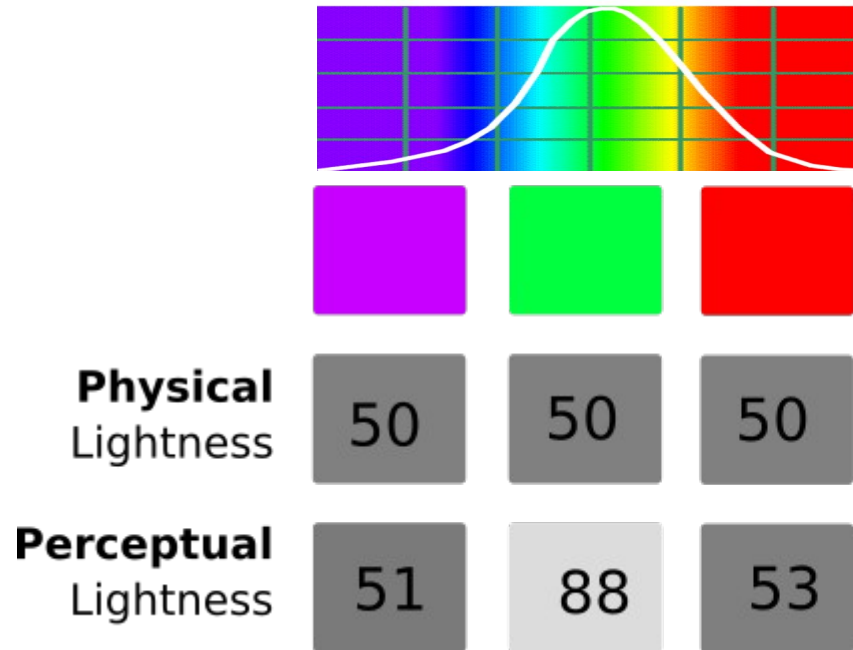
50

50

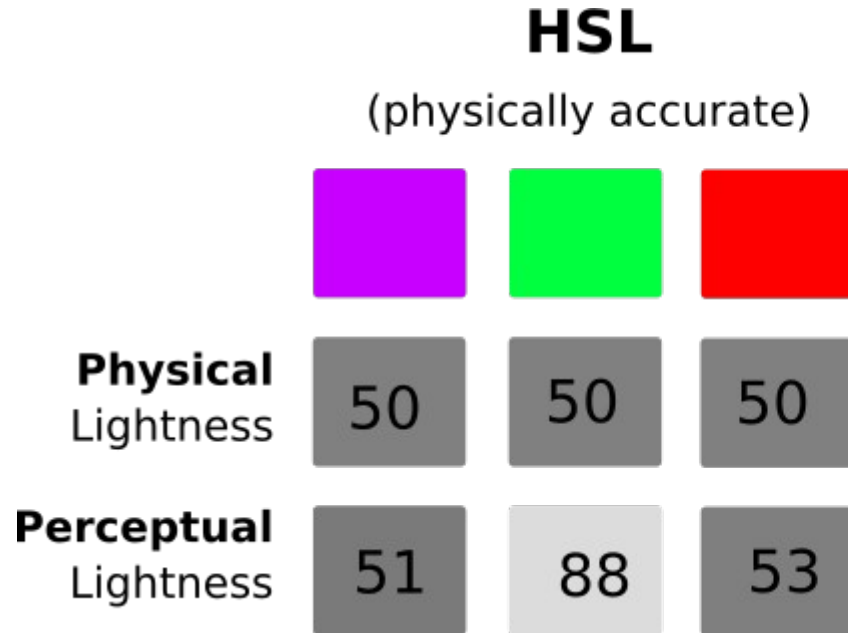
50

|                                |                                                                                   |                                                                                   |                                                                                    |
|--------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
|                                |  |  |  |
| <b>Physical</b><br>Lightness   |  |  |  |
| <b>Perceptual</b><br>Lightness |  |  |  |

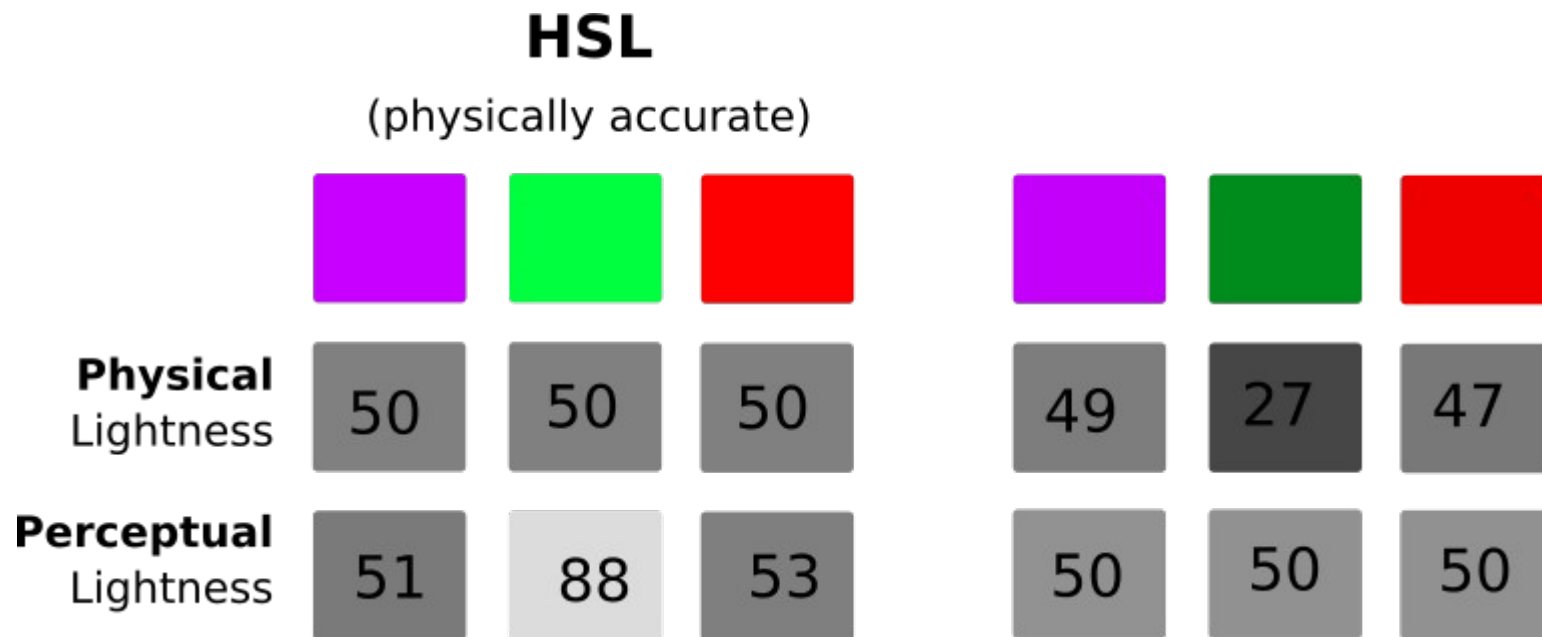
# Spectral sensitivity to luminance (lightness)



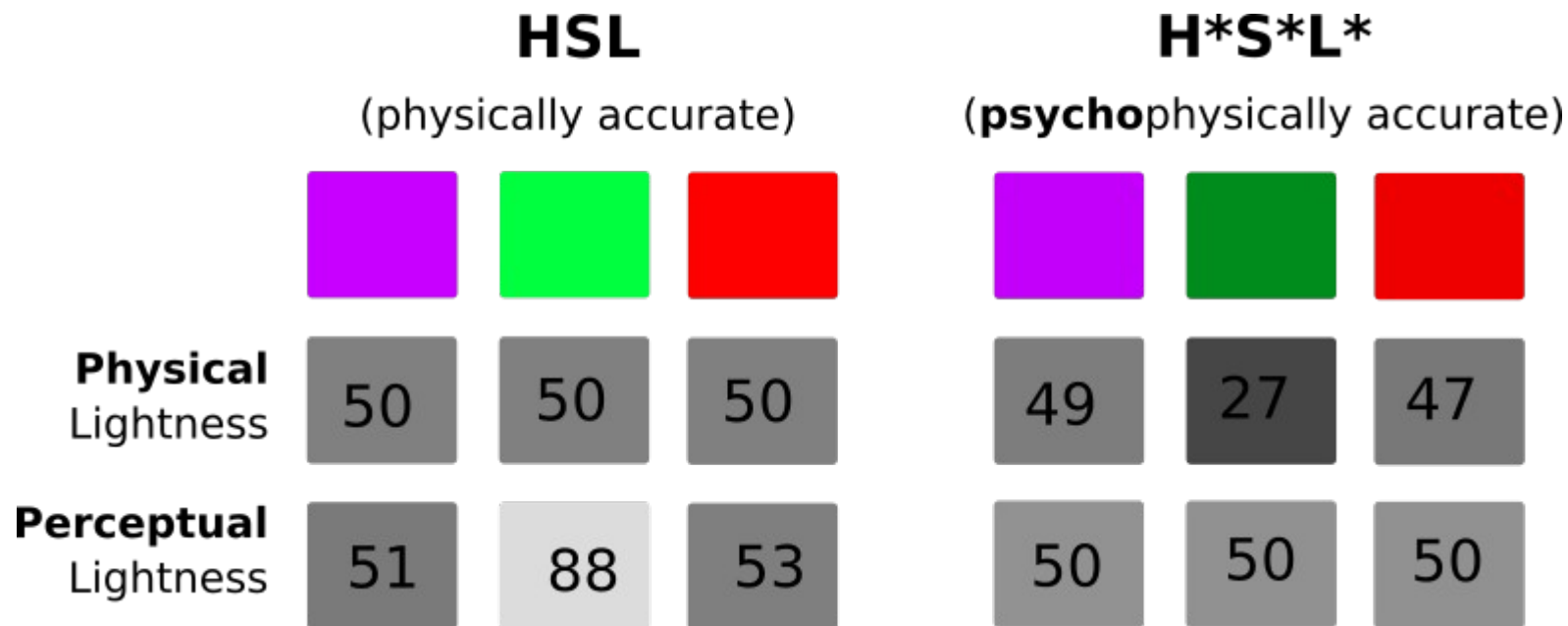
# Spectral sensitivity to luminance (lightness)



# Spectral sensitivity to luminance (lightness)



# Spectral sensitivity to luminance (lightness)



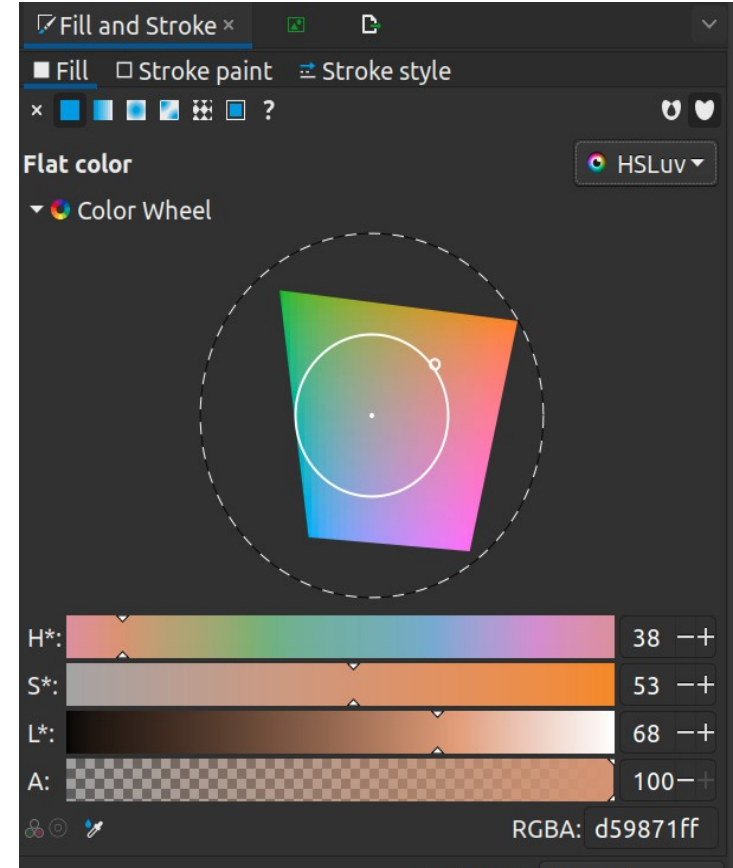
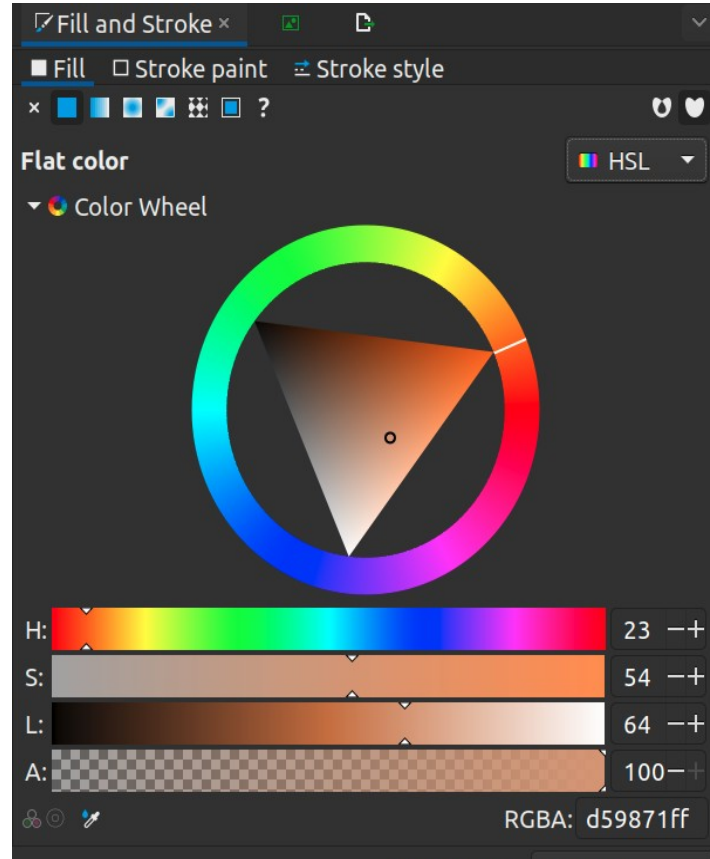
batlow



# HSL

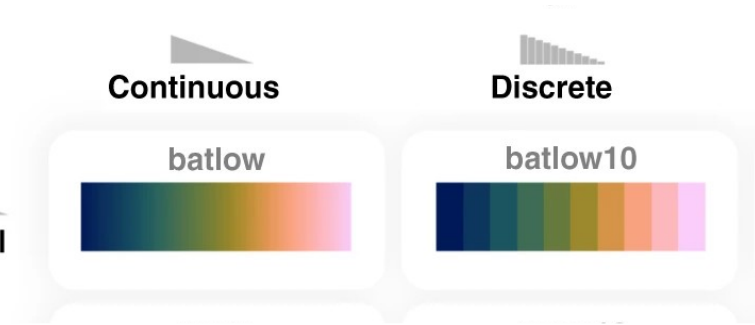
# vs H\*S\*L\* (or HSLuv)

Hue  
Saturation  
Lightness / Luminance



# Types of colormaps

Sequential



# Types of colormaps

Sequential

Diverging

Continuous

Discrete

batlow



batlow10



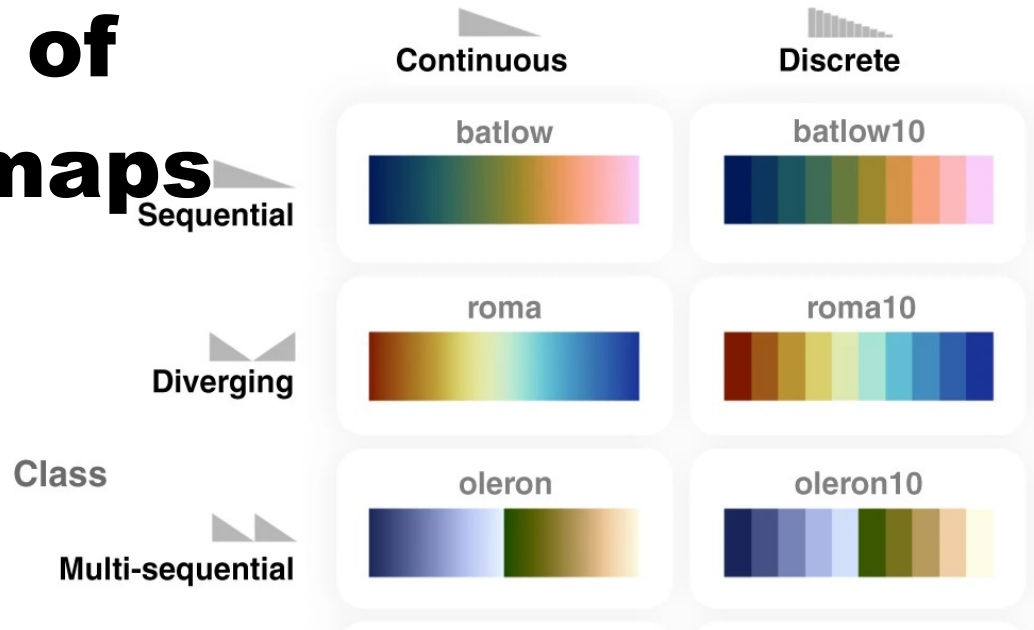
roma



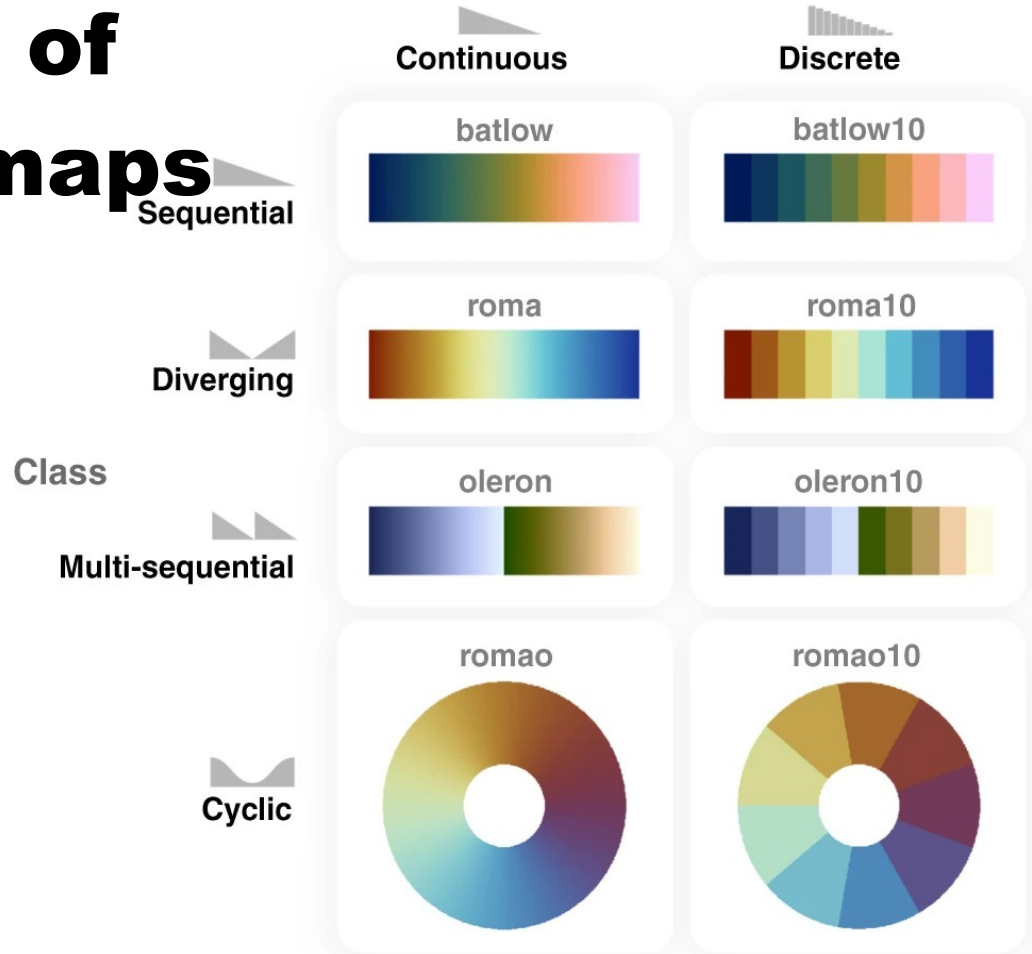
roma10



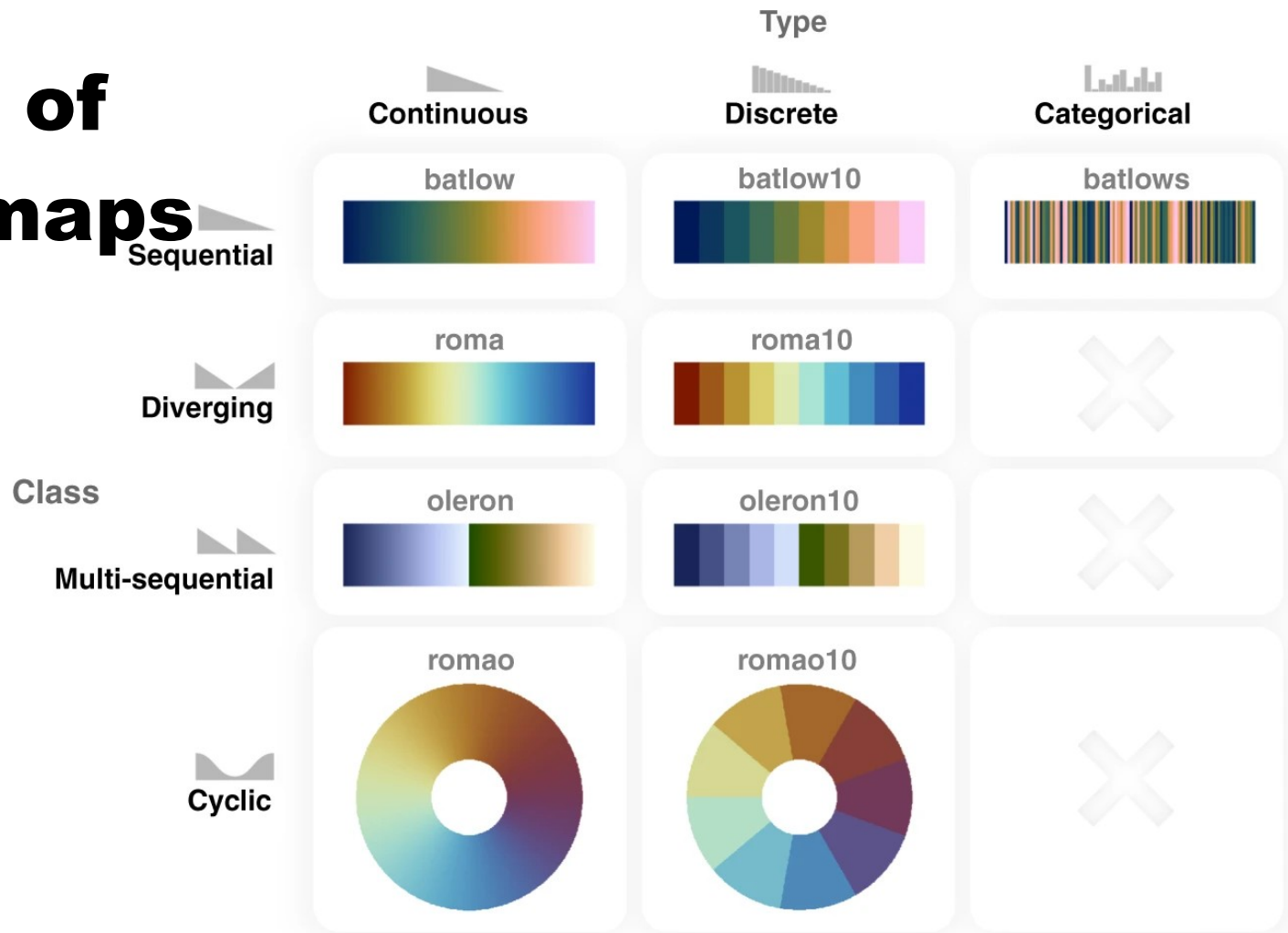
# Types of colormaps



# Types of colormaps



# Types of colormaps

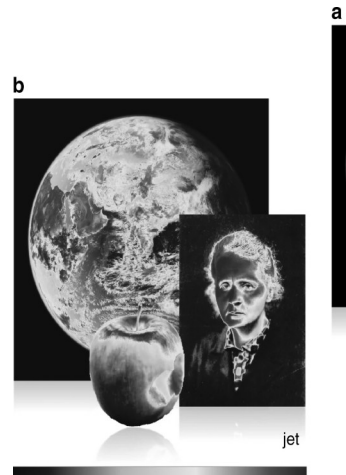


# How do I remember all that??

- Test yourself, your lab mate, your friend  
“Which bar is higher?”



- Look at the image in grayscale



(For future reference)

# Choosing effective charts



**Channels: Expressiveness Types and Effectiveness Ranks**

➔ **Magnitude Channels: Ordered Attributes**

|                             |  |                                                           |
|-----------------------------|--|-----------------------------------------------------------|
| Position on common scale    |  | <div>Most</div> <div>Effectiveness</div> <div>Least</div> |
| Position on unaligned scale |  |                                                           |
| Length (1D size)            |  |                                                           |
| Tilt/angle                  |  |                                                           |
| Area (2D size)              |  |                                                           |
| Depth (3D position)         |  |                                                           |
| Color luminance             |  |                                                           |
| Color saturation            |  |                                                           |
| Curvature                   |  |                                                           |
| Volume (3D size)            |  |                                                           |

➔ **Identity Channels: Categorical Attributes**

|                |  |
|----------------|--|
| Spatial region |  |
| Color hue      |  |
| Motion         |  |
| Shape          |  |
| Text           |  |

## Quick start

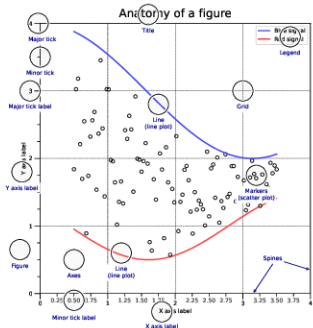
```
import numpy as np
import matplotlib as mpl
import matplotlib.pyplot as plt
```

```
X = np.linspace(0, 2*np.pi, 100)
Y = np.cos(X)
```

```
fig, ax = plt.subplots()
ax.plot(X, Y, color='green')
```

```
fig.savefig("Figure.pdf")
fig.show()
```

## Anatomy of a figure



## Subplots layout

```
subplot[s](rows,cols,...)
fig, axs = plt.subplots(3, 3)
```

```
G = gridspec(rows,cols,...)
ax = G[0,:]
```

```
ax.inset_axes(extent)
```

```
d=make_axes_locatable(ax)
ax = d.new_horizontal('10%')
```

## Getting help

- matplotlib.org
- github.com/matplotlib/matplotlib/issues
- discourse.matplotlib.org
- stackoverflow.com/questions/tagged/matplotlib
- https://git.io/matplotlib/matplotlib
- twitter.com/matplotlib
- Matplotlib users mailing list

## Basic plots

```
plot([X],[Y],[fmt],...)
```

```
scatter(X,Y,...)
X, Y, [s]izes, [c]olors, marker, cmap
```

```
bar[h](x,height,...)
x, height, width, bottom, align, color
```

```
imshow(Z,...)
Z, cmap, interpolation, extent, origin
```

```
contour[f]([X],[Y],Z,...)
X, Y, Z, levels, colors, extent, origin
```

```
pcolormesh([X],[Y],Z,...)
X, Y, Z, vmin, vmax, cmap
```

```
quiver([X],[Y],U,V,...)
X, Y, U, V, C, units, angles
```

```
pie(X,...)
Z, explode, labels, colors, radius
```

```
text(x,y,text,...)
x, y, text, va, ha, size, weight, transform
```

```
fill[_between][x](...)
```

## Advanced plots

```
step(X,Y,[fmt],...)
```

```
boxplot(X,...)
X, notch, sym, bootstrap, widths
```

```
errorbar(X,Y,xerr,yerr,...)
X, Y, xerr, yerr, fmt
```

```
hist(X, bins, ...)
```

```
violinplot(D,...)
D, positions, widths, vert
```

```
barbs([X],[Y], U, V, ...)
```

```
eventplot(positions,...)
positions, orientation, lineoffsets
```

```
hexbin(X,Y,C,...)
X, Y, C, gridsize, bins
```

## Scales

```
ax.set_[xy]scale(scale,...)
linear any values
symlog any values
log values > 0
logit 0 < values < 1
```

## Projections

```
subplot(...,projection=p)
p='polar'
p='3d'
p=ccrs.Orthographic()
import cartopy.crs as ccrs
```

## Lines

```
linestyle or ls
capstyle or dash_capstyle
```

```
markers
```

```
Colors
```

```
Colormaps
```

```
Ornaments
```

```
ax.legend(...)
```

```
ax.colorbar(...)
```

```
ax.annotate(...)
```

```
text
```

```
Event handling
```

```
fig, ax = plt.subplots()
def on_click(event):
    print(event)
fig.canvas.mpl_connect('button_press_event', on_click)
```

```
fig, ax = plt.subplots()
ax.set_[xy]scale(scale,...)
```

```
fig, ax = plt.subplots()
ax.set_[xy]scale(scale,...)
```

```
fig, ax = plt.subplots()
ax.set_[xy]scale(scale,...)
```

## Tick locators

```
from matplotlib import ticker
ax.[xy]axis.set_[minor|major]_locator(locator)
locator.NullLocator()
locator.MultipleLocator(0.5)
locator.FixedLocator([0, 1, 5])
locator.LinearLocator(numticks=3)
locator.IndexLocator(base=0.5, offset=0.25)
locator.AutoLocator()
locator.MaxNLocator(n=4)
locator.LogLocator(base=10, numticks=15)
```

## Tick formatters

```
from matplotlib import ticker
ax.[xy]axis.set_[minor|major]_formatter(formatter)
formatter.NullFormatter()
formatter.FixedFormatter(['zero', 'one', 'two', ...])
formatter.FuncFormatter(lambda x, pos: "[%2f] % x")
formatter.FormatStrFormatter("%>f")
formatter.ScalarFormatter()
formatter.StrMethodFormatter('%x')
formatter.PercentFormatter(xmax=5)
```

## Ornaments

```
ax.legend(...)
```

```
ax.colorbar(...)
```

```
ax.annotate(...)
```

```
text
```

```
Event handling
```

```
fig, ax = plt.subplots()
def on_click(event):
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```

```
fig, ax = plt.subplots()
ax.set_[xy]scale(scale,...)
```

```
fig, ax = plt.subplots()
ax.set_[xy]scale(scale,...)
```

```
fig, ax = plt.subplots()
ax.set_[xy]scale(scale,...)
```

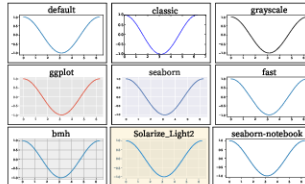
```
fig, ax = plt.subplots()
ax.set_[xy]scale(scale,...)
```

## Animation

```
import matplotlib.animation as mpla
T = np.linspace(0, 2*np.pi, 100)
S = np.sin(T)
line, = plt.plot(T, S)
def animate(i):
    line.set_ydata(np.sin(T+i/50))
anim = mpla.FuncAnimation(
    plt.gcf(), animate, interval=5)
plt.show()
```

## Styles

```
plt.style.use(style)
```



## Quick reminder

```
ax.grid()
ax.set_[xylim](vmin, vmax)
ax.set_[xy]label(label)
ax.set_[xy]ticks(ticks, [labels])
ax.set_[xy]ticklabels(labels)
ax.set_title(title)
ax.tick_params(width=10, ...)
ax.set_axis_[on|off]()
```

```
fig.suptitle(title)
fig.tight_layout()
plt.gcf(), plt.gca()
mpl.rc('axes', linewidth=1, ...)
[fig|ax].patch.set_alpha(0)
text=r'$\frac{-e^{\frac{1}{i}p}}{2^n}n$'
```

## Keyboard shortcuts

|        |                   |        |                   |
|--------|-------------------|--------|-------------------|
| Ctrl+S | Save              | Ctrl+W | Close plot        |
| r      | Reset view        | f      | Fullscreen 0/1    |
| f      | View forward      | b      | View back         |
| p      | Pan view          | o      | Zoom to rect      |
| x      | X pan/zoom        | y      | Y pan/zoom        |
| g      | Minor grid 0/1    | G      | Major grid 0/1    |
| l      | X axis log/linear | L      | Y axis log/linear |

## Ten simple rules

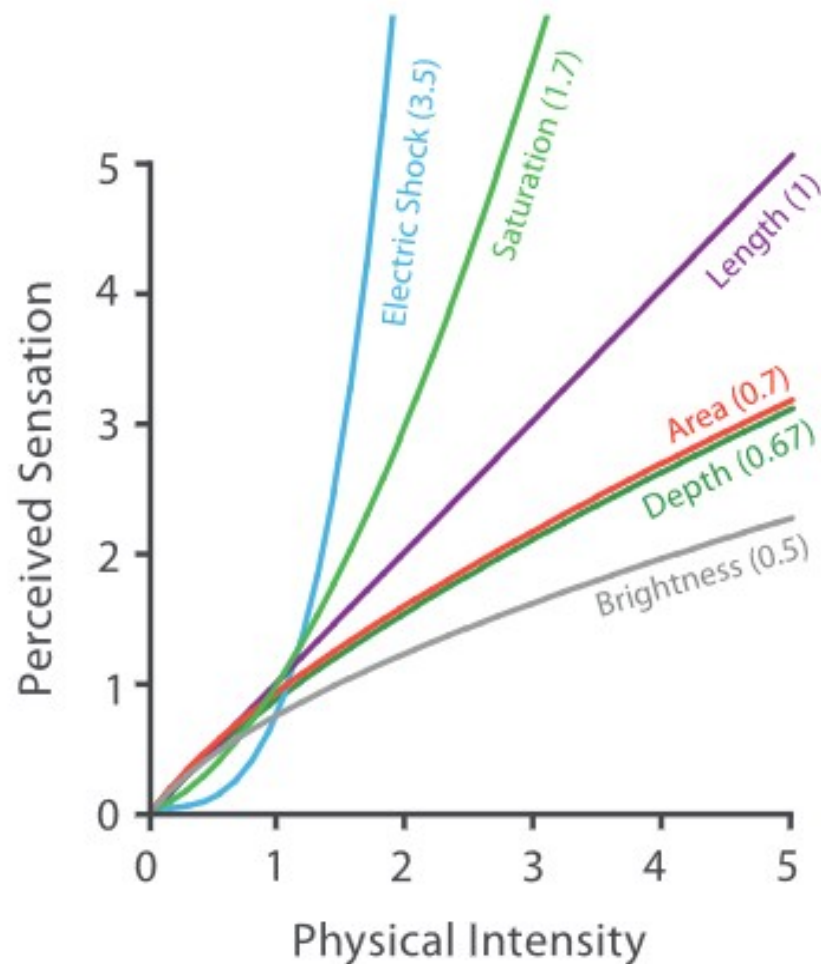
1. Know your audience
2. Identify your message
3. Adapt the figure
4. Captions are not optional
5. Do not trust the defaults
6. Use color effectively
7. Do not mislead the reader
8. Avoid "chartjunk"
9. Message trumps beauty
10. Get the right tool

# Lineplot

Ordered

Categorical

Steven's Psychophysical Power Law:  $S = I^N$



# Lineplot

## Ordered

Position on a  
common scale

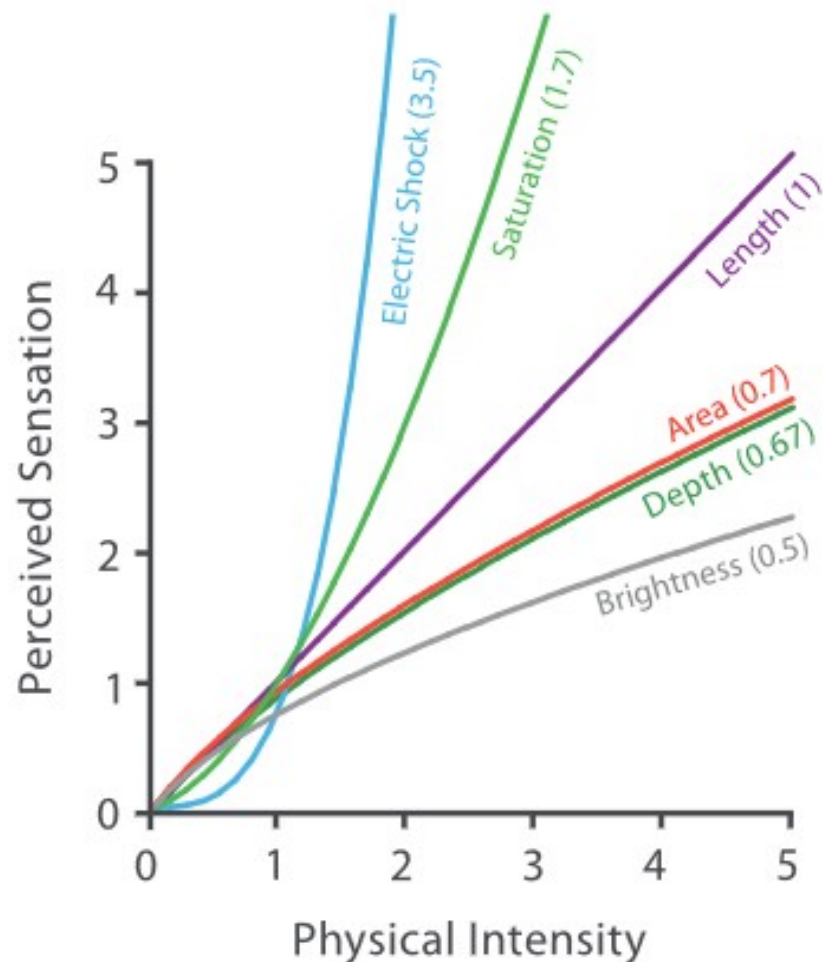


Text (equation)

Canada Canada Canada  
France France France

## Categorical

Steven's Psychophysical Power Law:  $S = I^N$



# Lineplot

## Ordered

Position on a  
common scale



Text (equation)

Canada Canada Canada  
France France France

## Categorical

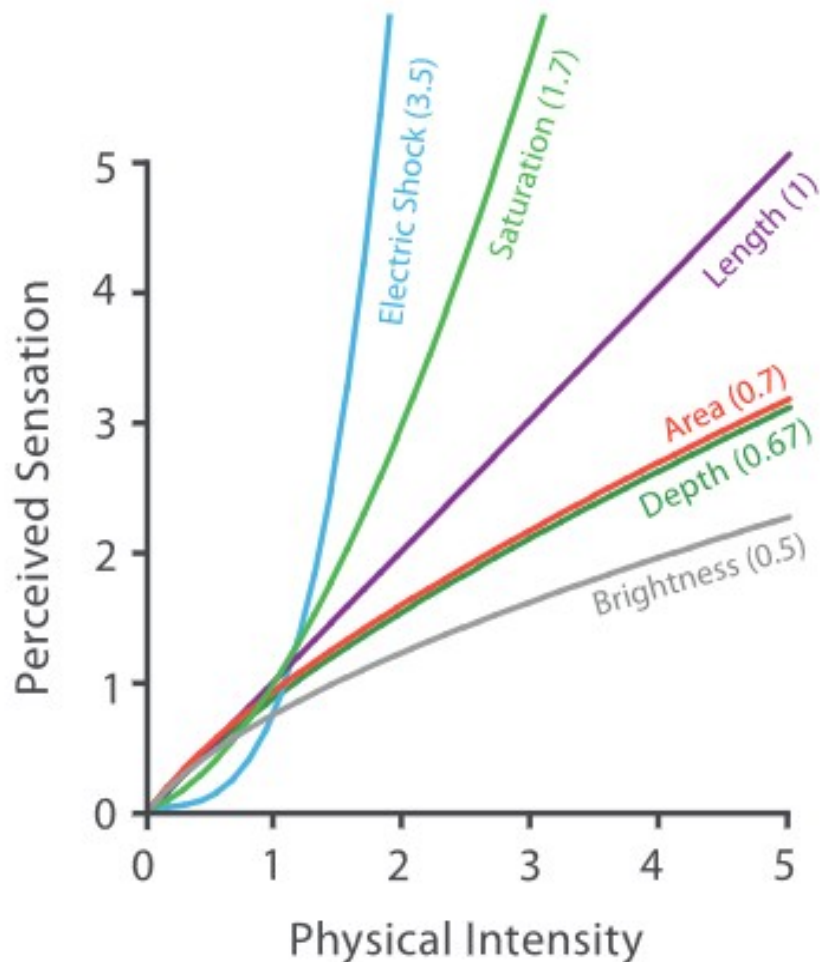
Hue



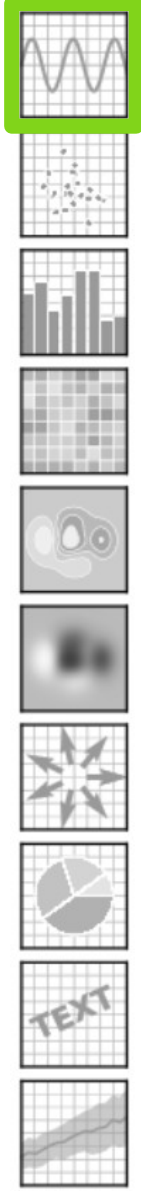
Text (identity)

Canada Canada Canada  
France France France

Steven's Psychophysical Power Law:  $S = I^N$

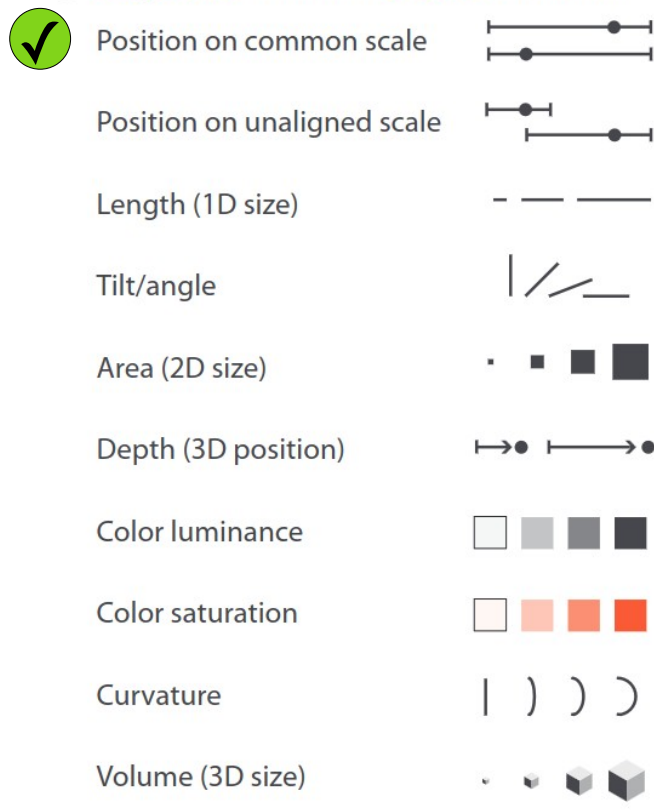


# Lineplot

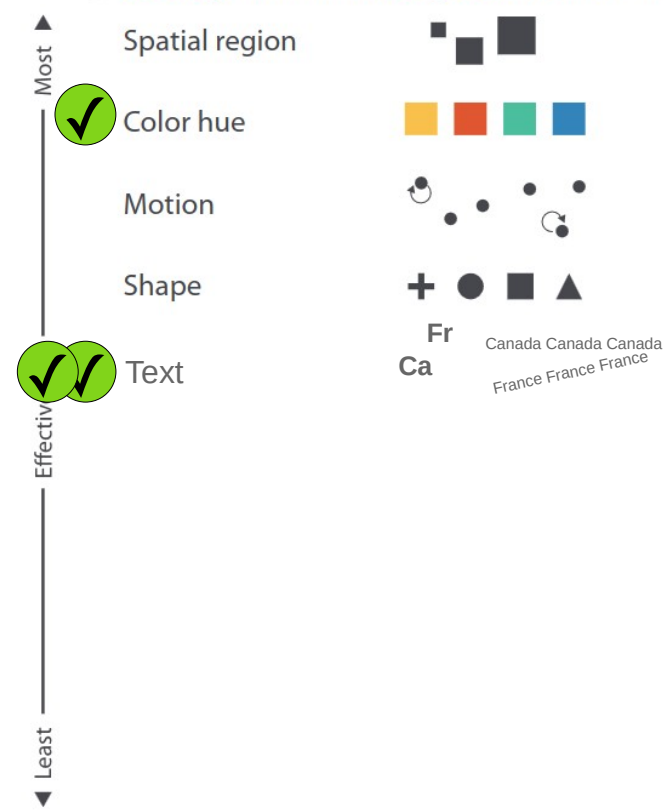


Channels: Expressiveness Types and Effectiveness Ranks

➔ **Magnitude Channels: Ordered Attributes**



➔ **Identity Channels: Categorical Attributes**

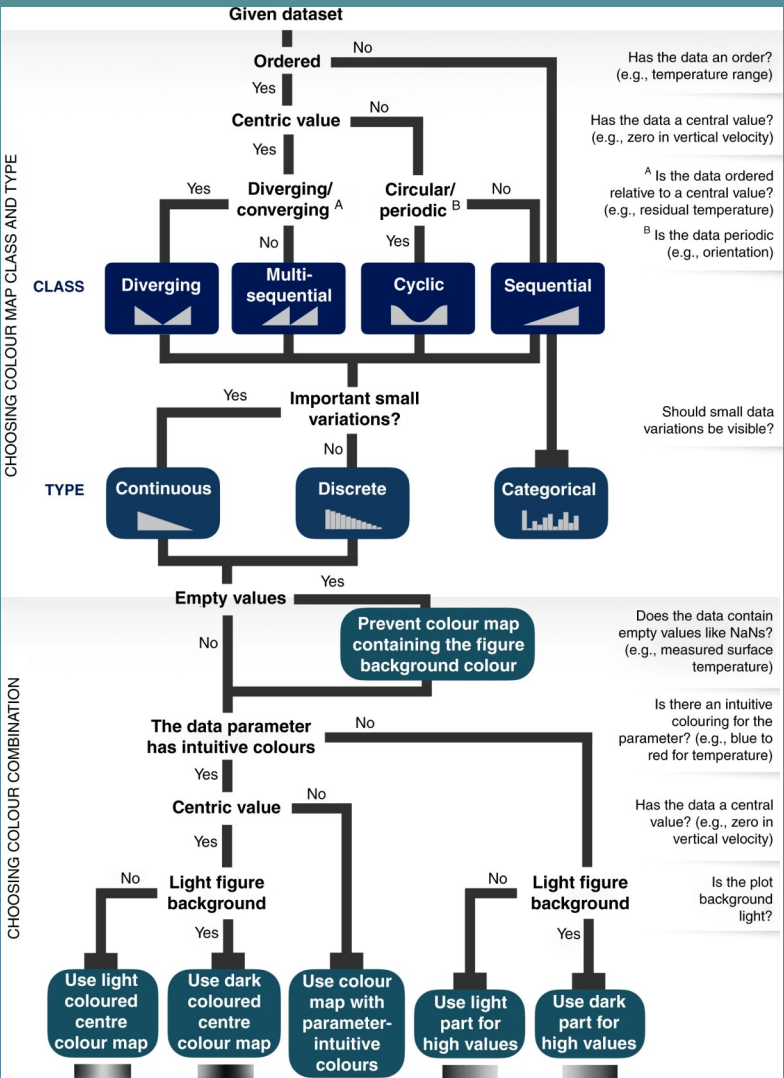


[https://matplotlib.org/cheatsheets/\\_images/cheatsheets-1.png](https://matplotlib.org/cheatsheets/_images/cheatsheets-1.png)

Munzner, T. (2014). Visualization analysis and design. CRC press.

(For future reference)

# Choosing effective colormaps



Activities

Firefox Web Browser

26

Frontiers

rogowitz a

IEEE Xplor

Data visual

IEEE Xplor

Fig. 2: Colo

Choosin x

https://matplotlib.org/stable/tutorials/colors/colormaps.html#sphx-gl-r-tutorials-colors-colorma

stable

matplotlib

Plot types Examples Tutorials Reference User guide Develop Releases

Section Navigation

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Customized Colorbars Tutorial

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Colormap Normalization

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Text

Toolkits

Provisional

Sequential

For the Sequential plots, the lightness value increases monotonically through the colormaps. This is good. Some of the  $L^*$  values in the colormaps span from 0 to 100 (binary and the other grayscale), and others start around  $L^* = 20$ . Those that have a smaller range of  $L^*$  will accordingly have a smaller perceptual range. Note also that the  $L^*$  function varies amongst the colormaps: some are approximately linear in  $L^*$  and others are more curved.

```
plot_color_gradients('Perceptually Uniform Sequential',  
                    ['viridis', 'plasma', 'inferno', 'magma', 'cividis'])
```

Perceptually Uniform Sequential colormaps

viridis

plasma

inferno

magma

cividis

```
plot_color_gradients('Sequential',  
                    ['Greys', 'Purples', 'Blues', 'Greens', 'Oranges', 'Re  
                    'YlOrRd', 'OrRd', 'PuRd', 'RdPu', 'BuPu',
```

On this page

Overview

Classes of colormaps

Sequential

Sequential2

Diverging

Cyclic

Qualitative

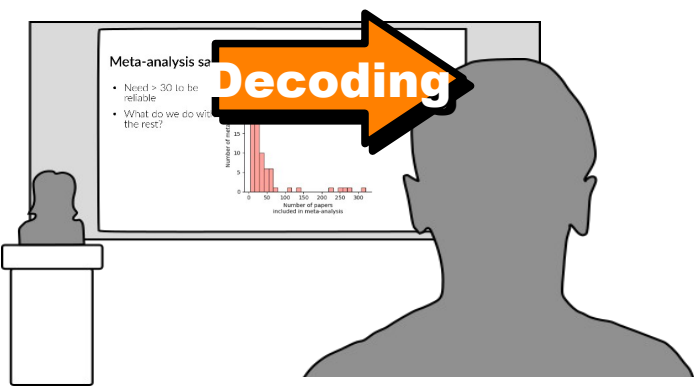
Miscellaneous

Lightness of Matplotlib colormaps

Grayscale conversion

Color vision deficiencies

References



To plan an **effective visualization**, we need to think about

- **Message**

- **What** we want to communicate

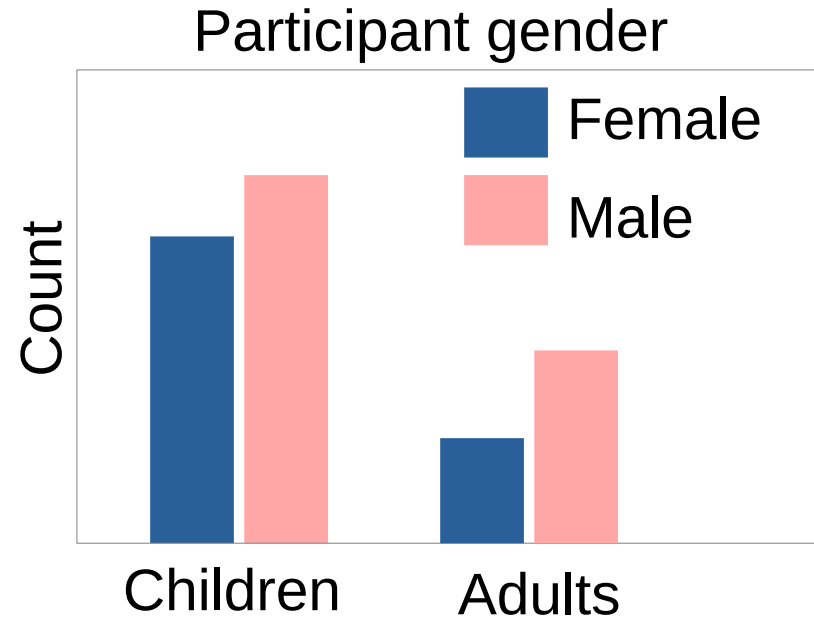
- **Perception**

- **How best** to communicate it

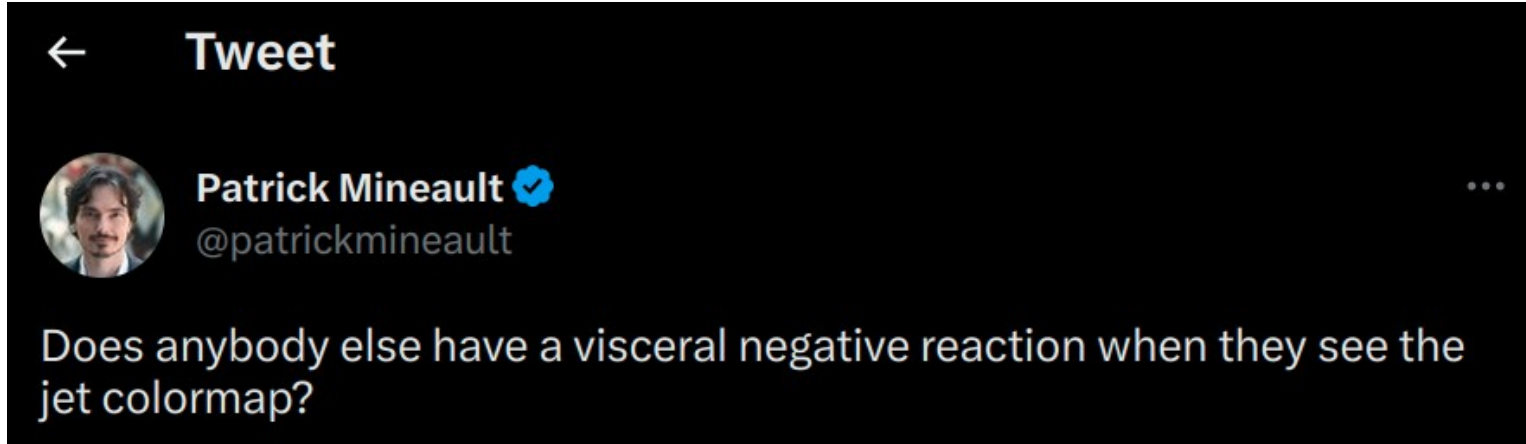
- **Conventions**

- **How** it's **usually** communicated

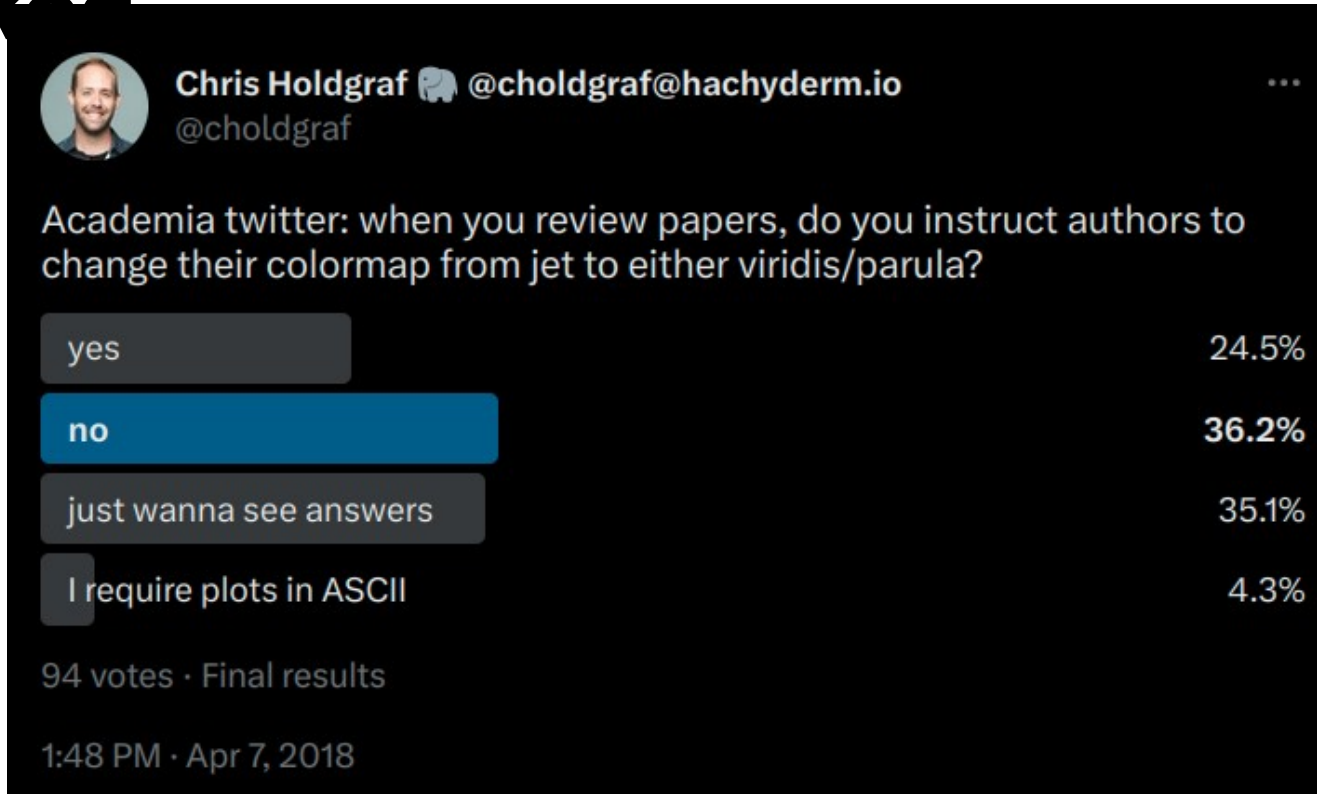
# Culture: Color associations



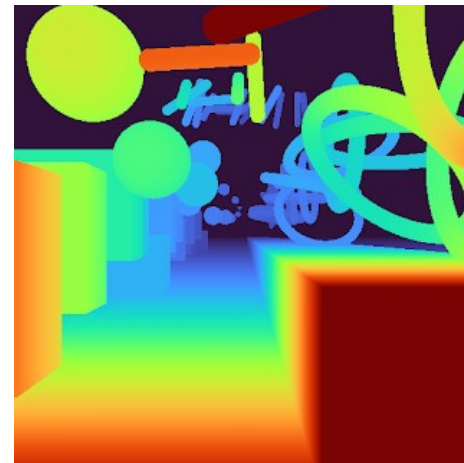
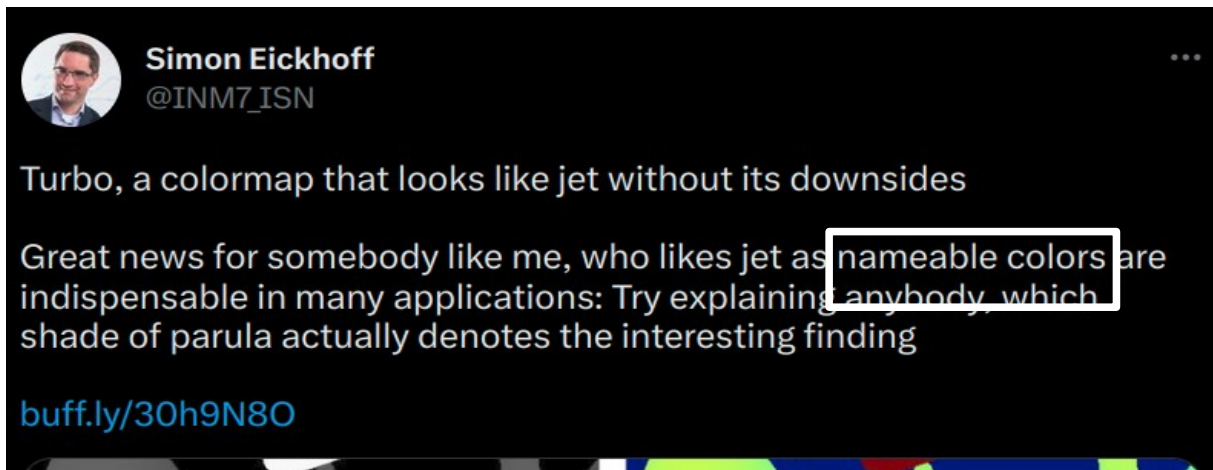
# Science: Rainbow colormaps are disliked



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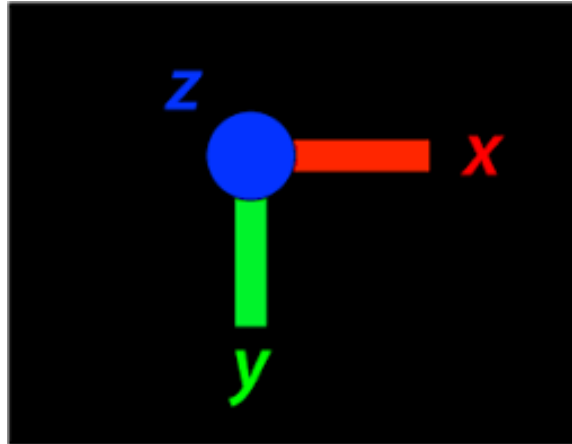
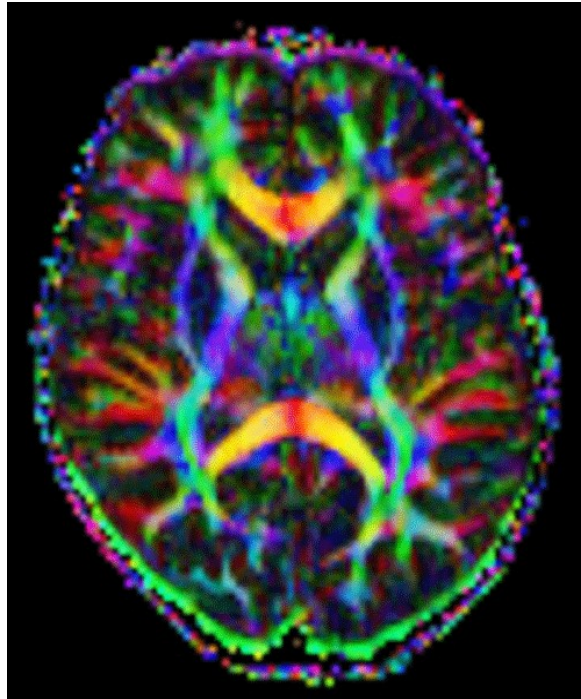


# Science: Rainbow colormaps are disliked?



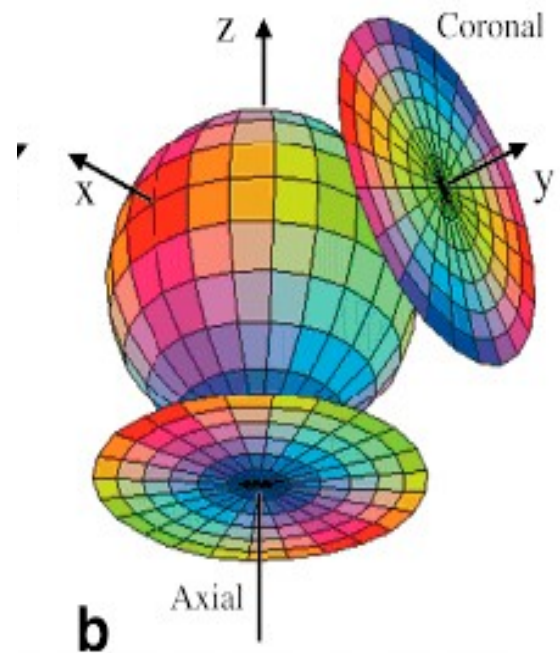
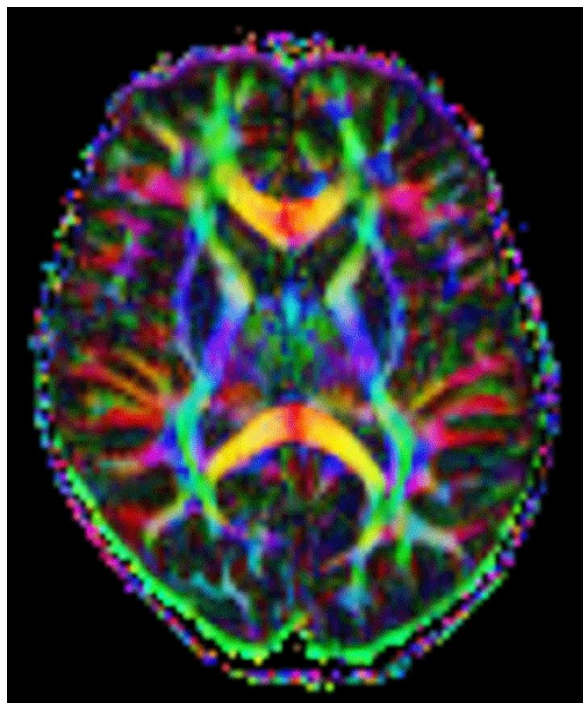
# Neuroscience: Colors = 3D direction

Principal Diffusion Direction



# Neuroscience: Colors = 3D direction

Principal Diffusion Direction

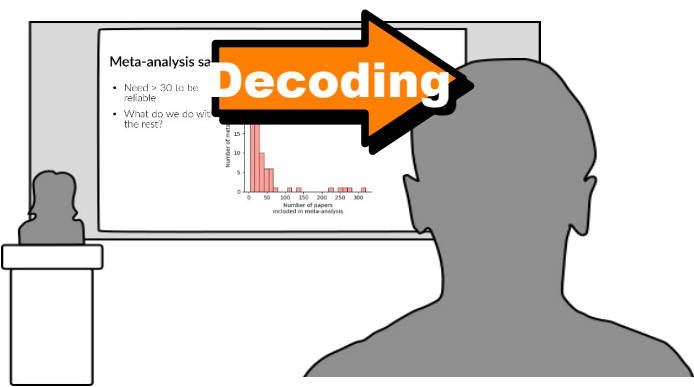


(For future reference)

# Examples of visualization

## conventions

- Culture
  - Pink is female
  - Time goes left to right
- Science
  - Rainbow colormaps are disliked
- Neuroscience
  - MRI in grayscale
  - DWI colors = directions



To plan an **effective visualization**, we need to think about

- **Message**

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- **How** it's **usually** communicated

# More self-explanatory

## Meta-analysis sample sizes

- Need > 30 to be reliable
- What do we do with the rest?

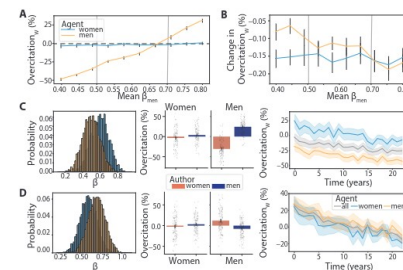
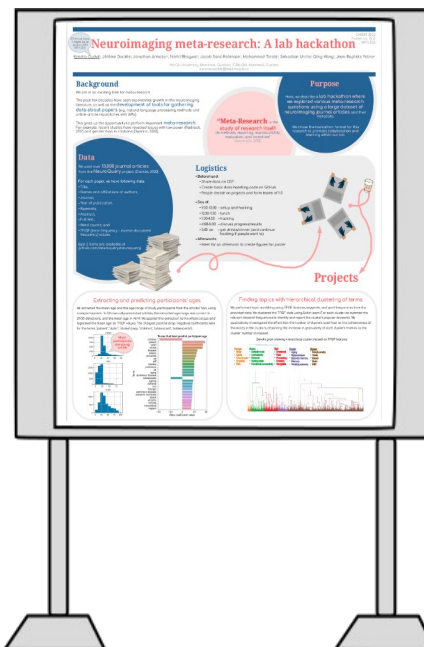
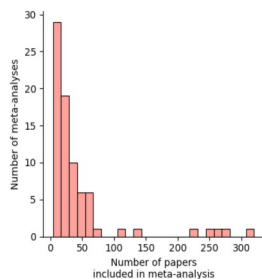


FIG. 3. **Interventions to network sampling** ( $\beta$ ). (A) The total overcitation of women for women (blue) and men (yellow) agents as the mean value of men's  $\beta$  changes. Grey lines indicate parameter values that are shown in panels (C) and (D). (B) The same information as in panel (A), but visualizing the slope of overcitation of women rather than the total overcitation. (C) The total (center) and time varying (right) overcitation when men's  $\beta$  values are centered on 0.5 (left). (D) The total (center) and time varying (right) overcitation when men's  $\beta$  values are centered on 0.7 (left).

later (1) the average overcitation of women and (2) the average slope of overcitation over time. These two calculations allow us to assess whether each intervention is addressing citation disparities, their worsening over time, or both. Overall, we find that increasing the  $\beta$  value of men increases the overcitation of women (Fig. 3A), but does not reduce, and actually strengthens, the citation deficit over time (Fig. 3B). This stronger citation deficit could be due to a floor effect, where agents with the lowest citations of women overall have little room to decline. We also note that a  $\beta$  value of around 0.5, which indicates no bias in citations or discussions, is not enough to ensure equitable citations (Fig. 3C;  $t$ -test  $t(256) = -25.5, p = 2.57 \times 10^{-5}$ ). The smallest  $\gamma$  where equitable citations in men arise is around 0.05, which corresponds to a preference for citing and discussing women (Fig. 3D;  $t$ -test  $t(256) = 2.89, p = 4.57 \times 10^{-3}$ ). This simulation demonstrates that increasing the likelihood of discussing and citing women's scholarship above their representation in the field leads to more equitable citation practices, but does not reverse temporal trends.

### Interventions to biased meeting preferences

We next simulated the effects of intervention to the meeting preferences of men. Here, we select 10 values for  $\gamma$  that range between 0.001 and 0.06 (the original value used). The value of 0.001 effectively means that agents will learn from all other agents, even if they share no authors in their citation histories. Broadly, we demonstrate that increasing the selectivity of agents that men learn from decreases an already negative overcitation of women (Fig. 3A), and leads to sharper decreases in the overcitation of women over time (Fig. 3B). We next examine a single value of  $\gamma$  that produced the most equitable citation practices (Fig. 3C; 0.01). We find that for  $\gamma$  values of 0.01, men agents still significantly under-

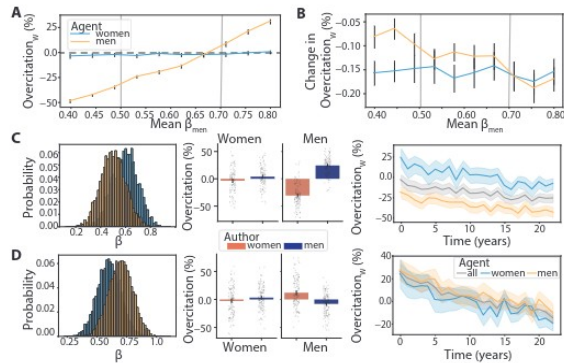
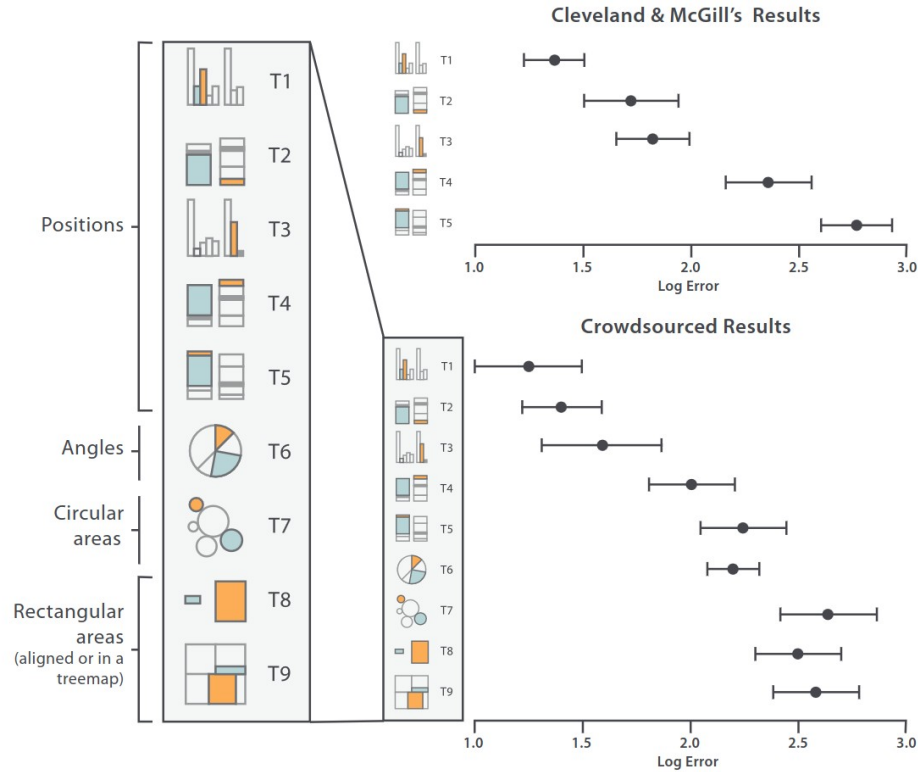


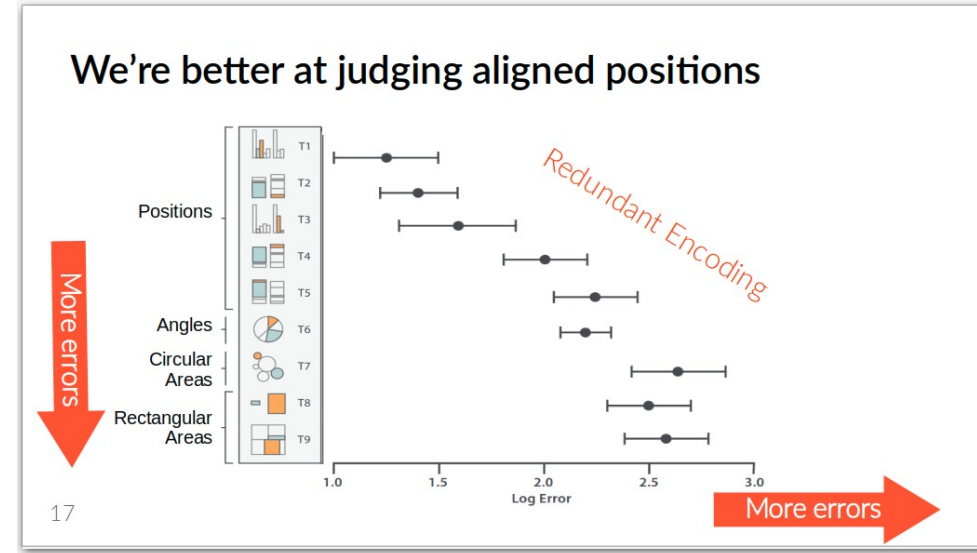
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- How long would it take them to understand this slide?
- How long will they see the slide?

E.g., Figure in textbook

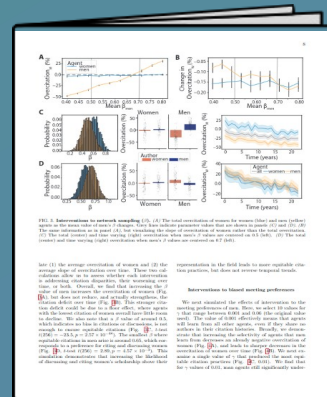
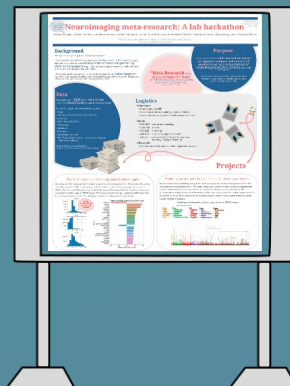
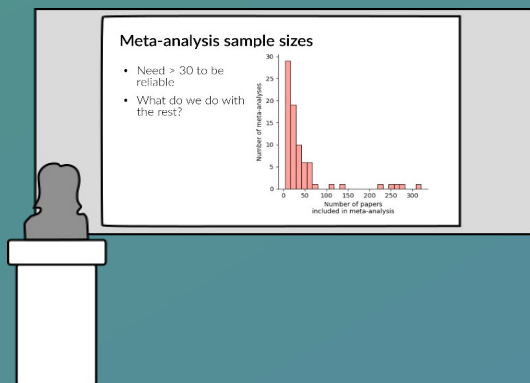


My slide



(For future reference)

# Contextual adjustments



Time to understand

Proximity to viz

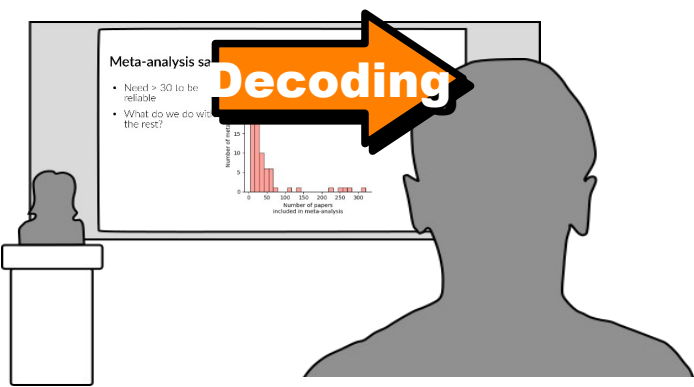
Text explanations

Self-explanatory

Visual complexity

Less

More



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# Example 1 – Simple data and figure

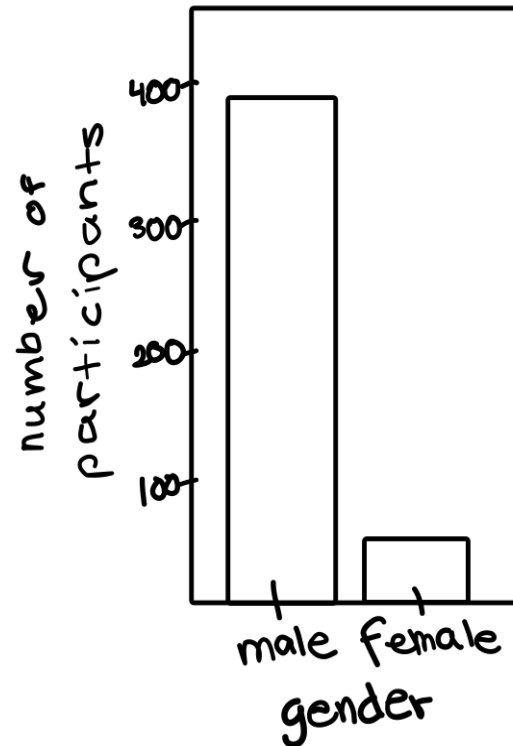
- Original paper on the ABIDE dataset
    - 964 subjects
      - 396 male
      - 51 female
- ← **Lets visualize this**

# What's our message?

- More male than female participants
  - 2 categories, 2 values

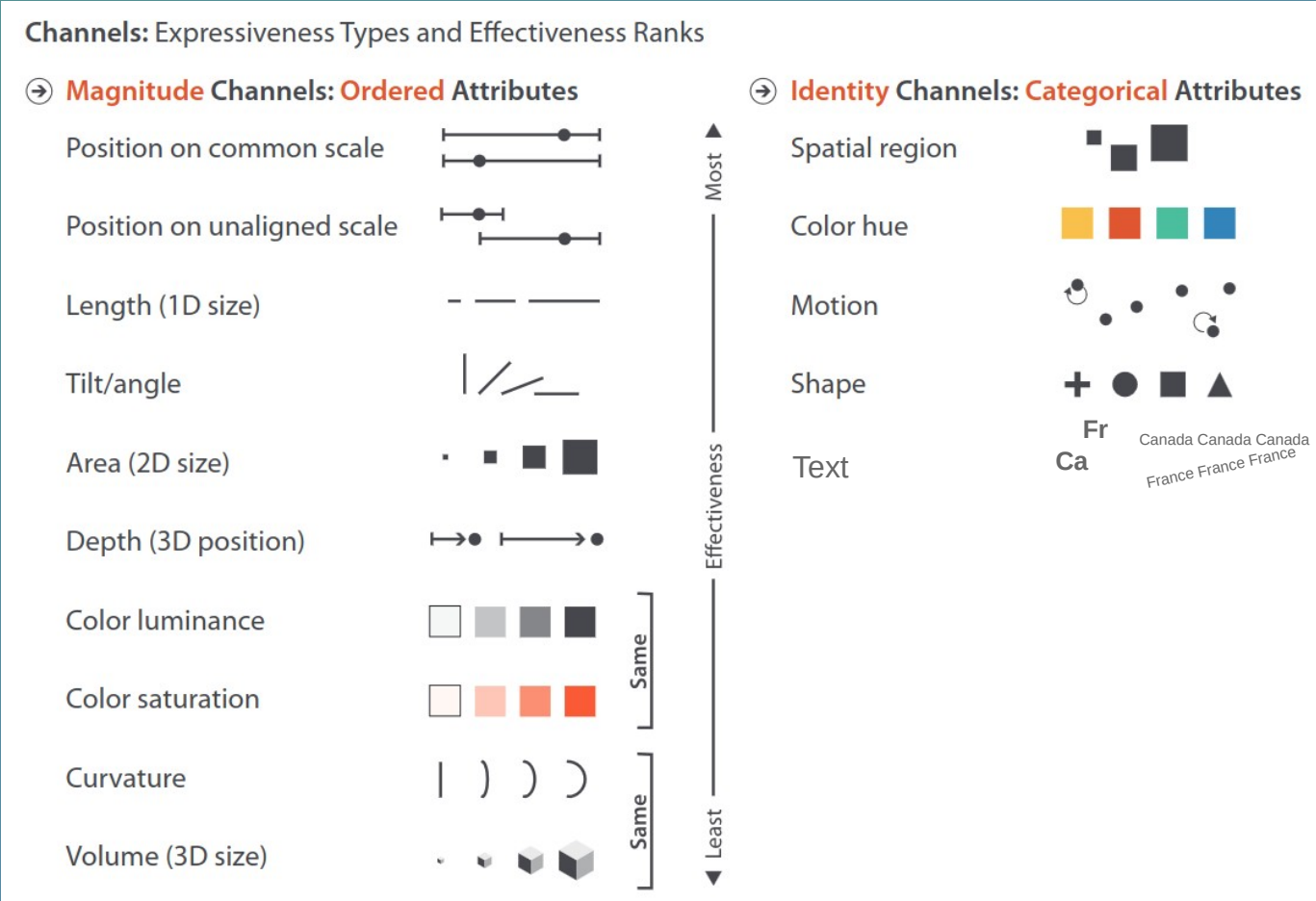
# How should we communicate this message?

- Clearly show 2 categories and 2 different values



(For future reference)

# Choosing effective charts

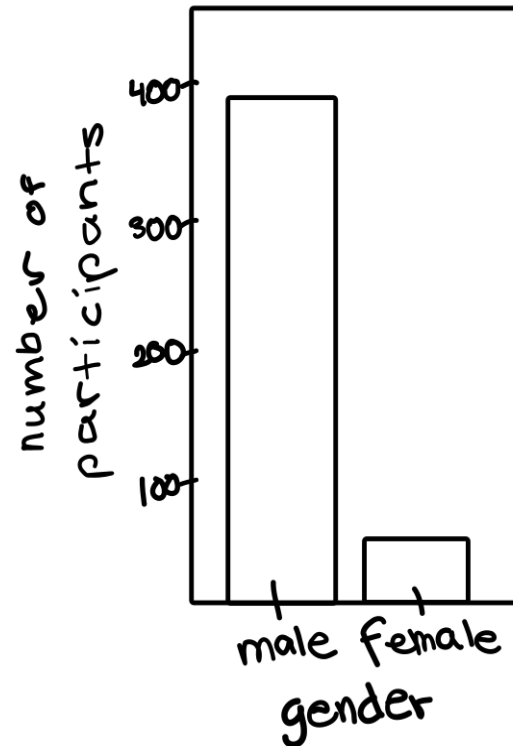


[https://matplotlib.org/cheatsheets/\\_images/cheatsheets-1.png](https://matplotlib.org/cheatsheets/_images/cheatsheets-1.png)

Munzner, T. (2014). Visualization analysis and design. CRC press.

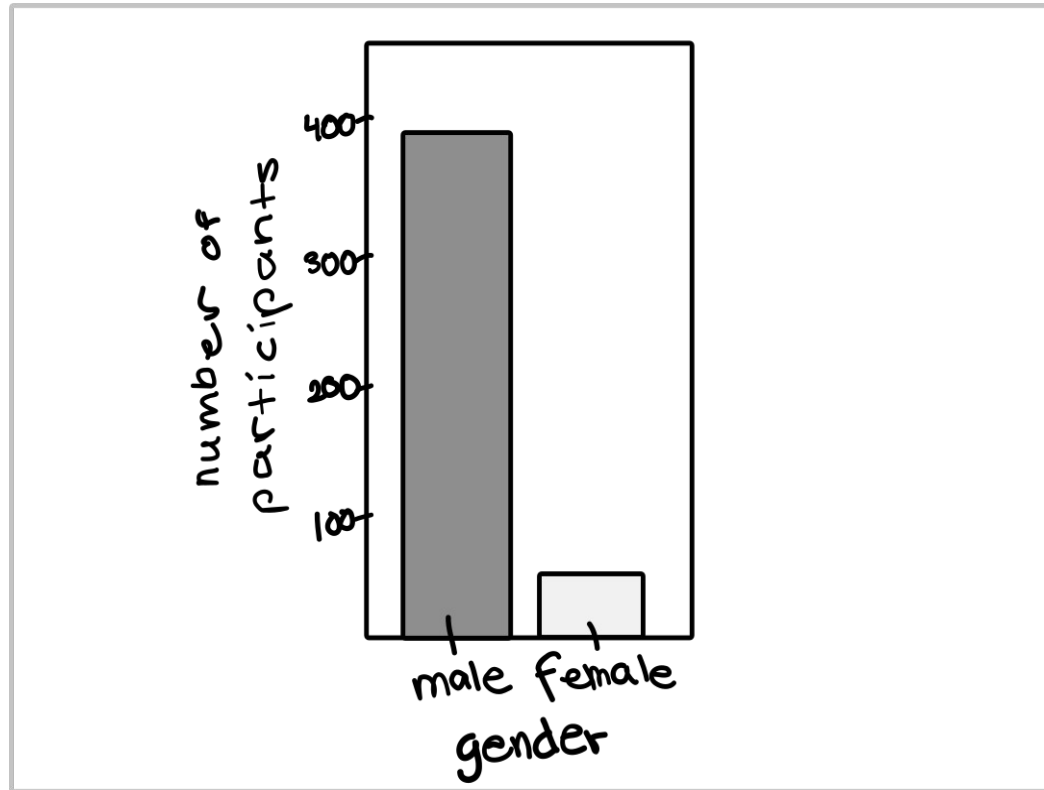
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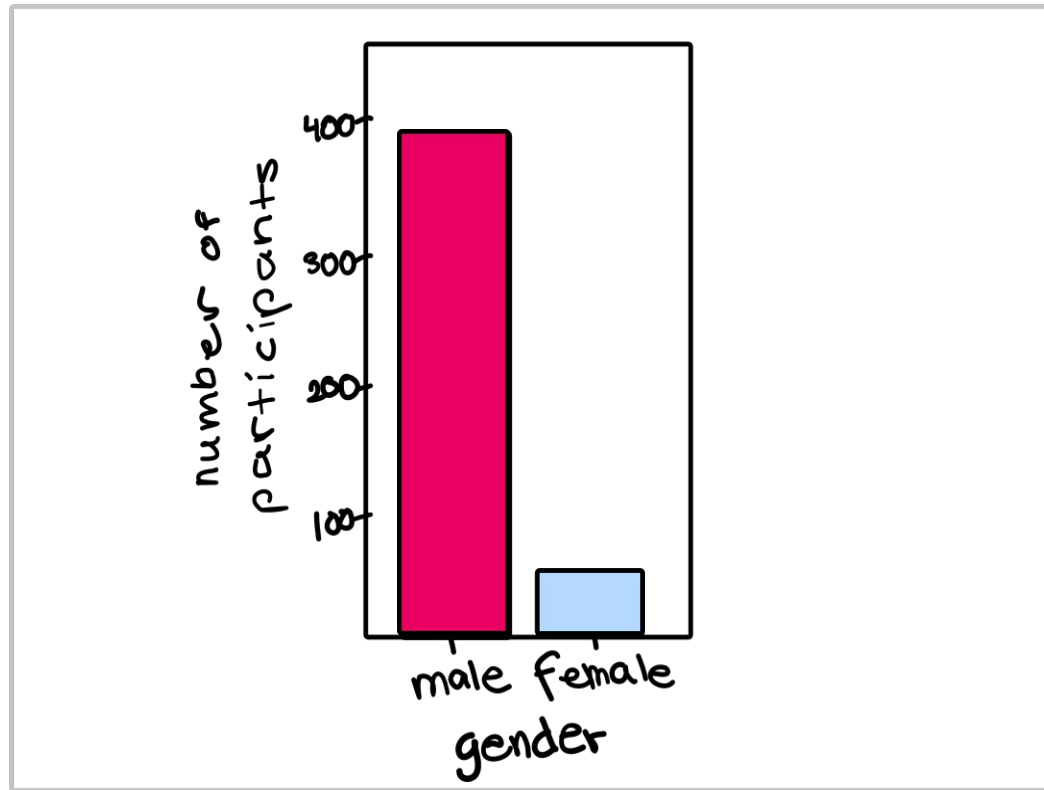
# How should we communicate this message?

- Clearly show 2 categories and 2 different **values**



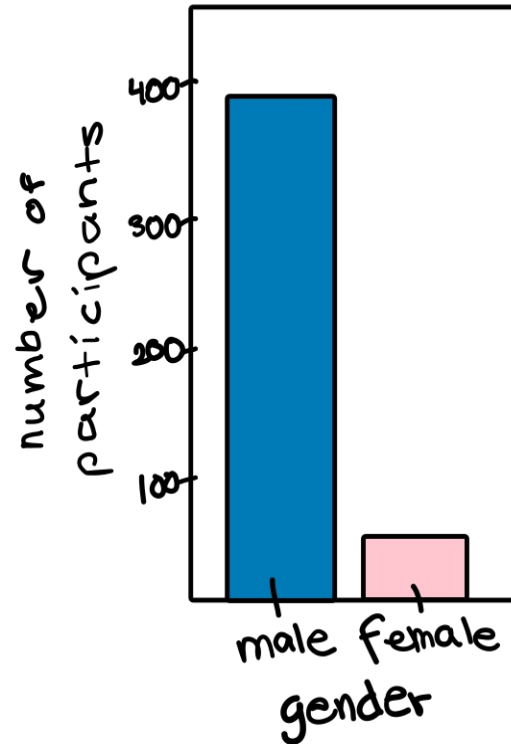
# How should we communicate this message?

- Clearly show 2 **categories** and 2 different values



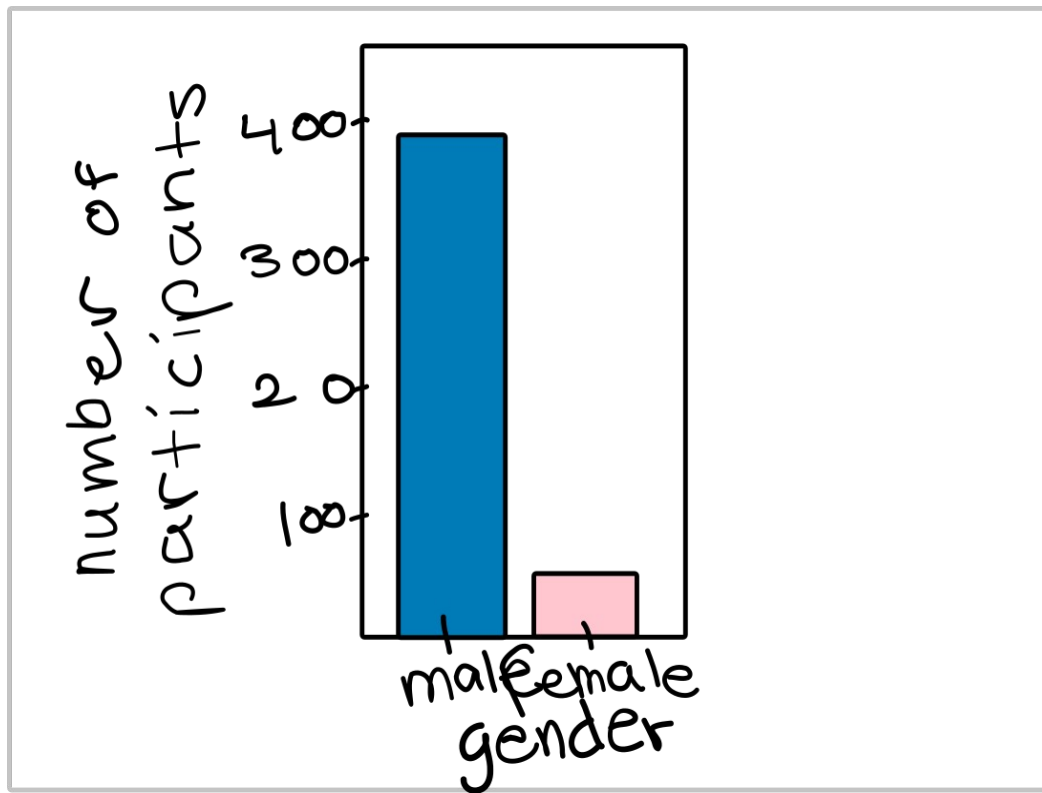
# Are there any conventions we should think about?

- Gender & color



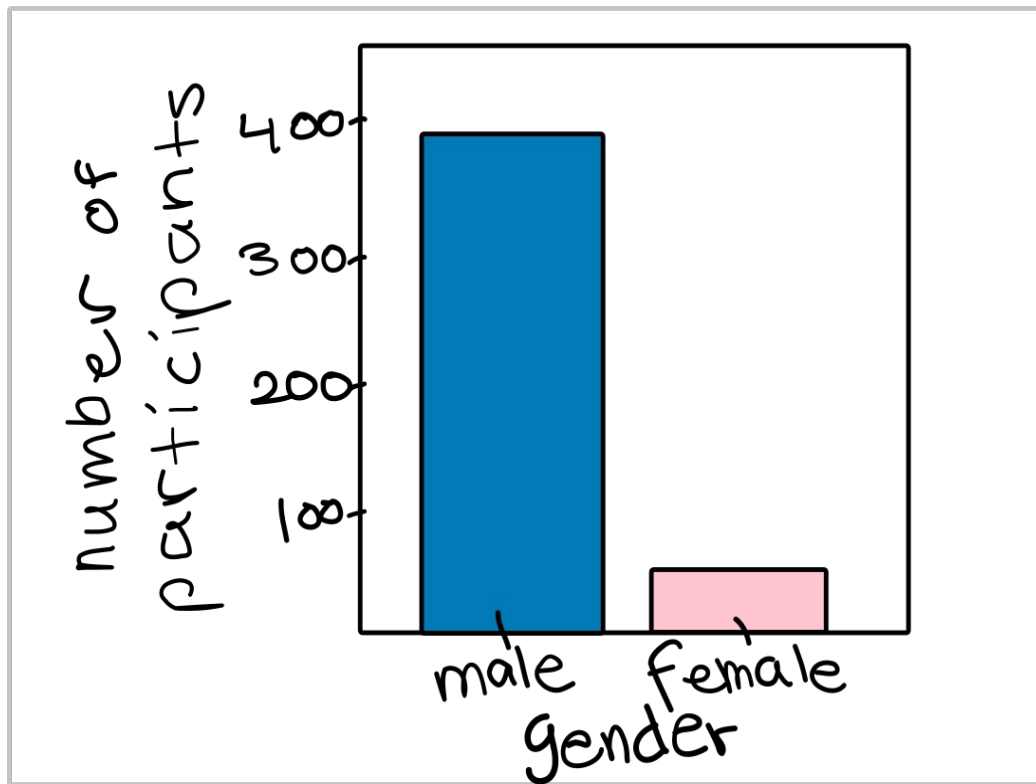
# Do we need to adapt it to the context?

- Academic presentation



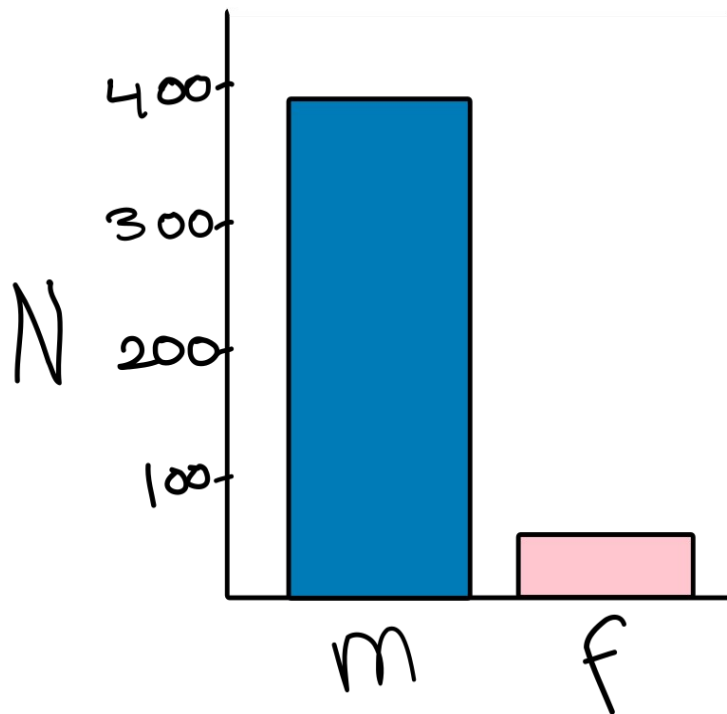
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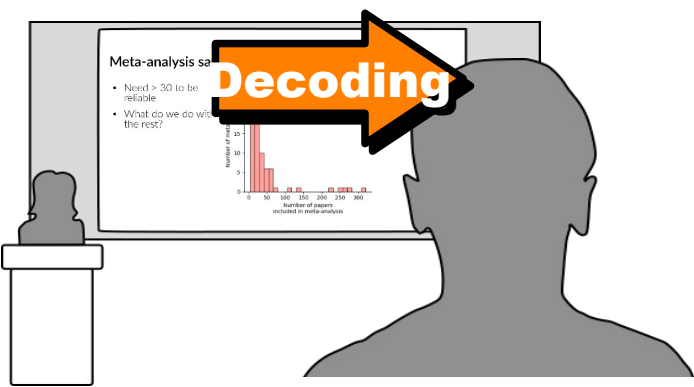
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# The end of part 1

# Reference slides...

(For future reference)

# Choosing effective charts



### Channels: Expressiveness Types and Effectiveness Ranks

#### ➔ Magnitude Channels: Ordered Attributes

|                          |  |                                                           |
|--------------------------|--|-----------------------------------------------------------|
| Position on common scale |  | <div>Most</div> <div>Effectiveness</div> <div>Least</div> |
|--------------------------|--|-----------------------------------------------------------|

| Position on unaligned scale |  |
| Length (1D size) |  |
| Tilt/angle |  |
| Area (2D size) |  |
| Depth (3D position) |  |
| Color luminance |  |
| Color saturation |  |
| Curvature |  |
| Volume (3D size) |  |

#### ➔ Identity Channels: Categorical Attributes

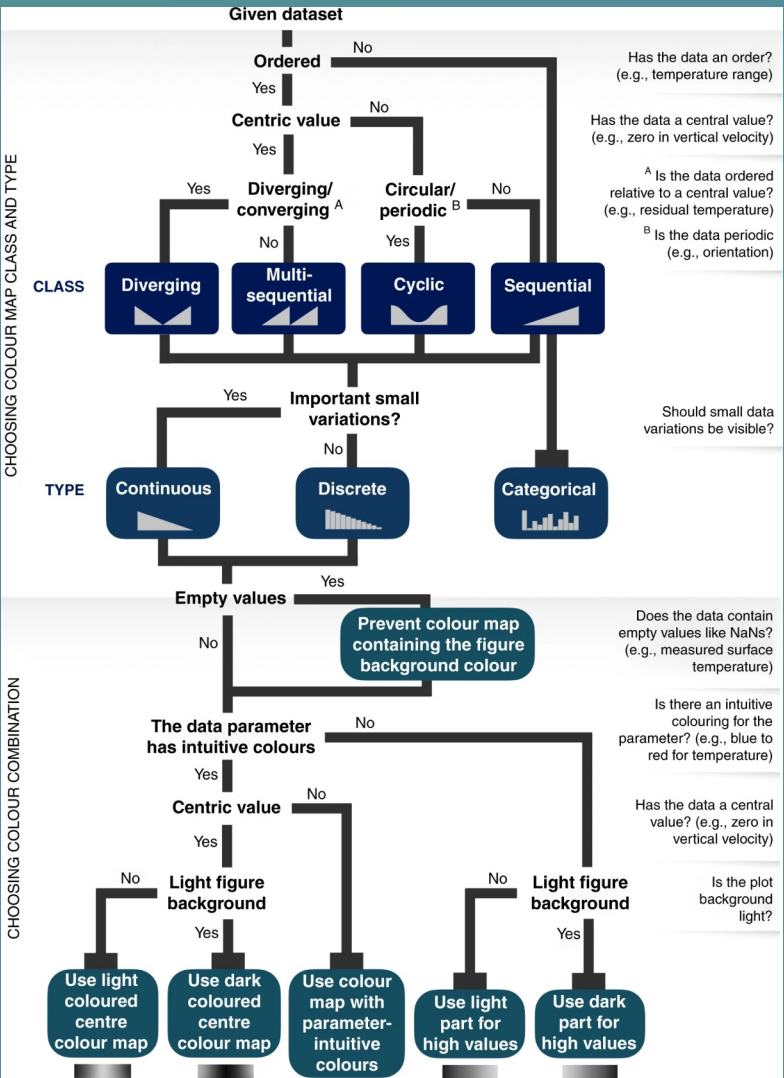
|                |  |
|----------------|--|
| Spatial region |  |
| Color hue      |  |
| Motion         |  |
| Shape          |  |
| Text           |  |

[https://matplotlib.org/cheatsheets/\\_images/cheatsheets-1.png](https://matplotlib.org/cheatsheets/_images/cheatsheets-1.png)

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(For future reference)

# Choosing effective colormaps



(For future reference)

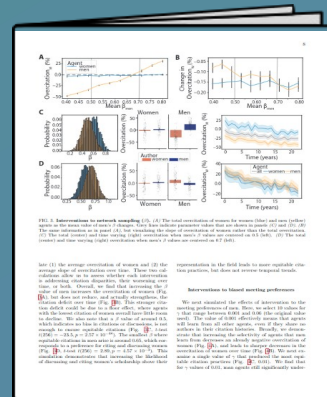
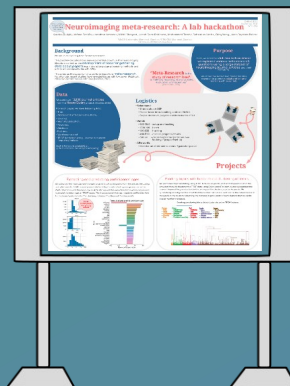
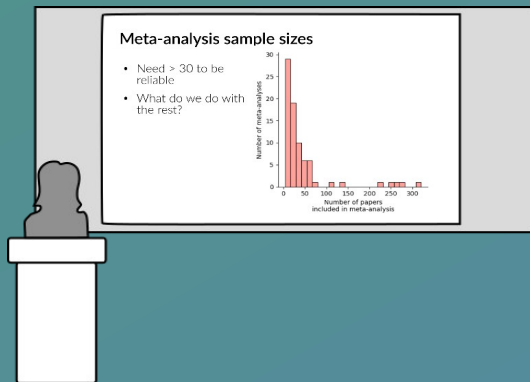
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